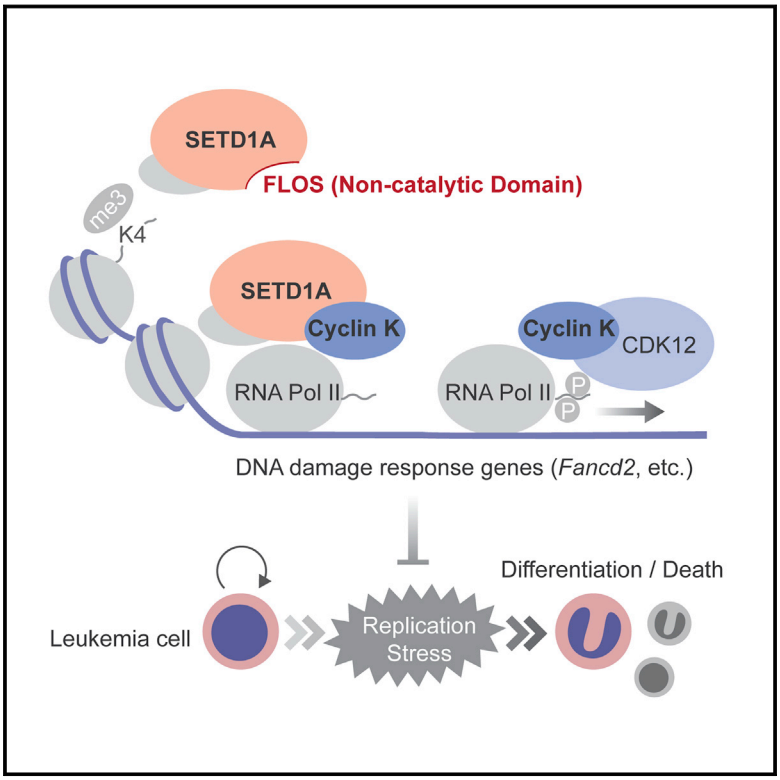
Article

Cell

A Non-catalytic Function of SETD1A Regulates Cyclin K and the DNA Damage Response

Graphical Abstract



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In Brief

Independent of its enzymatic activity, H3K4 methyltransferase SETD1A promotes leukemic cell survival by regulating DNA damage response.

Highlights

- H3K4 methyltransferase SETD1A has non-redundant functions with SETD1B
- Non-enzymatic domain of SETD1A is indispensable for leukemia cell growth
- The SETD1A FLOS domain regulates expression of genes involved in DNA damage response
- The SETD1A FLOS domain interacts with cyclin K to regulate gene expression

Hoshii et al., 2018, Cell *172*, 1007–1021
February 22, 2018 © 2018 Elsevier Inc.
https://doi.org/10.1016/j.cell.2018.01.032

CellPress

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A Non-catalytic Function of SETD1A Regulates Cyclin K and the DNA Damage Response

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https://doi.org/10.1016/j.cell.2018.01.032

SUMMARY

MLL/SET methyltransferases catalyze methylation of histone 3 lysine 4 and play critical roles in development and cancer. We assessed MLL/SET proteins and found that SETD1A is required for survival of acute myeloid leukemia (AML) cells. Mutagenesis studies and CRISPR-Cas9 domain screening show the enzymatic SET domain is not necessary for AML cell survival but that a newly identified region termed the "FLOS" (functional location on SETD1A) domain is indispensable. FLOS disruption suppresses DNA damage response genes and induces p53-dependent apoptosis. The FLOS domain acts as a cyclin-K-binding site that is required for chromosomal recruitment of cyclin K and for DNA-repair-associated gene expression in S phase. These data identify a connection between the chromatin regulator SETD1A and the DNA damage response that is independent of histone methylation and suggests that targeting SETD1A and cyclin K complexes may represent a therapeutic opportunity for AML and, potentially, for other cancers.

INTRODUCTION

Histone 3 lysine 4 methylation (H3K4me) is a post-translational modification associated with transcriptional regulation in multiple organisms from yeast to humans (Barski et al., 2007; Briggs et al., 2001; Ng et al., 2003; Shilatifard, 2008). The enzymes that catalyze this modification are important for the maintenance of developmental gene expression in multicellular organisms and are frequently mutated in human cancer (Bledau et al., 2014; Denissov et al., 2014; Ernst et al., 2004; Kandoth et al., 2013). H3K4 histone methyltransferases (HMTs) are defined by an enzymatic SET domain (Gu et al., 1992; Jones and Gelbart, 1993; Tschiersch et al., 1994). Yeast have a single H3K4 HMT, SET1, that is present in a multiprotein (COMPASS) complex responsible for all H3K4 methylation (Schneider et al., 2005). Mammals have 6 homologs of yeast SET1 that possess H3K4 HMT activity, including MLL1–4 (KMT2A–D) and SETD1A/B (KMT2F/G). Each of these enzymes is found in a multiprotein complex with overlapping subunits. While each of these complexes have been shown to possess H3K4 HMT activity, the extent to which their cellular functions are largely a result of SET-domain-associated enzymatic activity is unclear.

One of the most well-characterized mammalian SET1 homologs is MLL1. MLL1 is found in fusion proteins generated by chromosomal translocation of the *MLL1* gene to one of more than 60 different partner genes (Krivtsov and Armstrong, 2007). Pediatric acute leukemia cases with *MLL1*-rearrangement (*MLL*-r) have a poor prognosis (Balgobind et al., 2009). Recent studies have shown that wild-type MLL proteins play a role in *MLL*-r leukemia, but the mechanistic details and whether these roles are dependent on H3K4 HMT activity are just beginning to be explored (Milne et al., 2010; Santos et al., 2014; Thiel et al., 2010; Wong et al., 2015). Wild-type MLL1 is also indispensable for several subtypes of acute myeloid leukemia (AML) beyond *MLL*-r AML including MN1-expressing leukemia, *NPM1*-mutated leukemia and *NUP98*-rearranged leukemia (Kühn et al., 2016; Riedel et al., 2016; Xu et al., 2016). However, the catalytic SET domain of the MLL1 protein is dispensable for leukemia growth suggesting a non-catalytic function of wild-type MLL1 and redundancies such that other SET1 homologs might contribute H3K4 HMT activity (Kühn et al., 2016; Mishra et al., 2014). Since the enzymatic domains of the SET1 vertebrate homologs are highly conserved during evolution, these homologs might have both redundant and unique functions in regulating H3K4 methylation and leukemia cell proliferation. Distinguishing the dependencies on enzymatic and non-enzymatic functions in leukemia will shape small-molecule drug development focused on targeting chromatin-associated, epigenetic mechanisms.

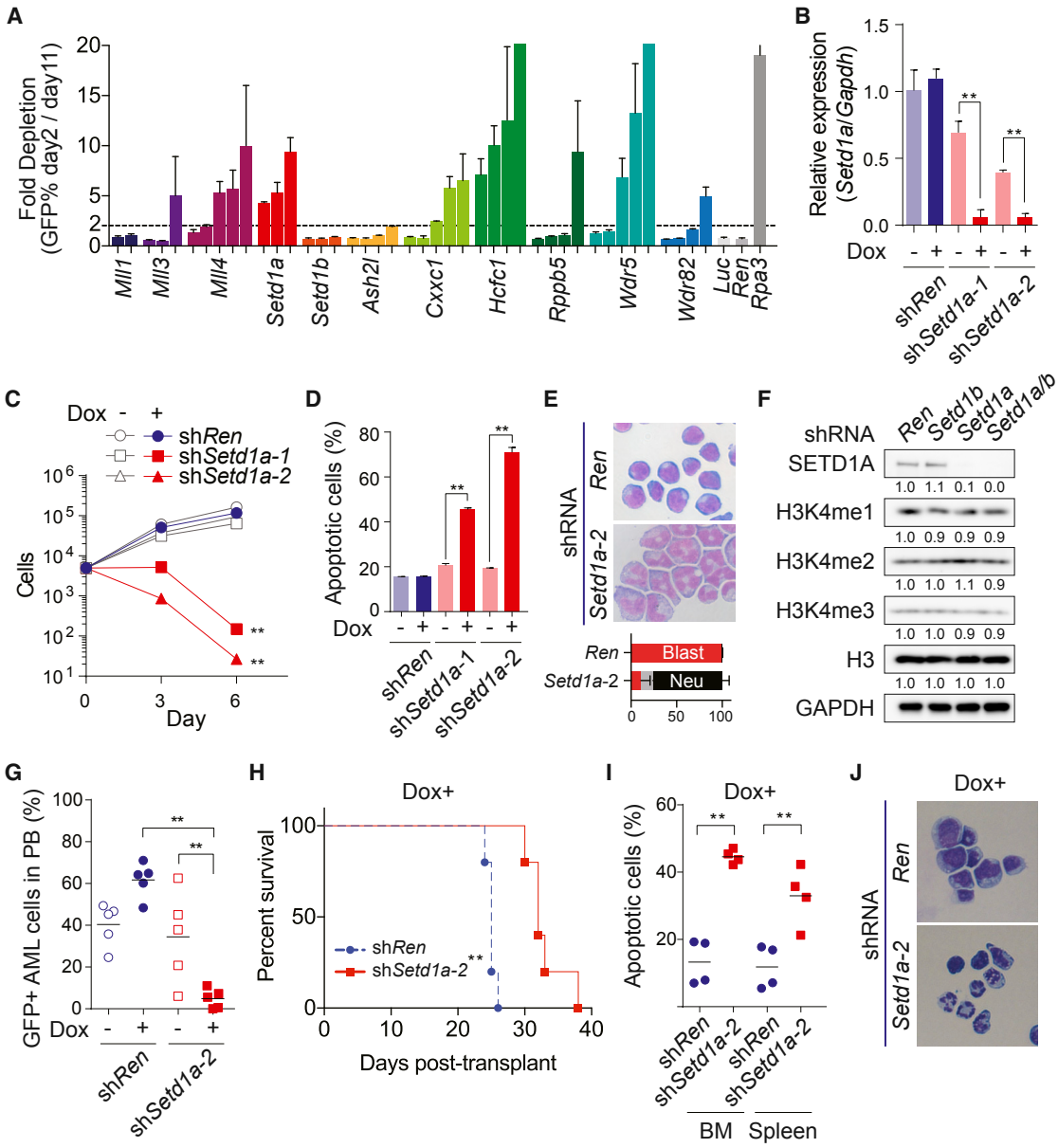


Figure 1. SET/COMPASS Complex Is Required for Leukemia Cell Survival and H3K4me3 Modification

(A) The indicated MLL/SET/COMPASS complex subunit shRNAs with a GFP reporter were expressed in mouse MLL-AF9 leukemia cells. This experiment was repeated 2 times with 3 biological replicates in each experiment.

(B) *Setd1a* expression was analyzed in mouse MLL-AF9 leukemia cells that were transfected with doxycycline (dox)-inducible *Renilla luciferase* (sh*Ren*) control or *Setd1a* shRNAs (sh*Setd1a*-1 and sh*Setd1a*-2). Cells were analyzed at 4 days post-dox. Representative data from 1 out of 3 independent experiments with 3 biological replicates are shown.

(C) Numbers of dox-inducible *Ren* or *Setd1a* shRNA-expressing cells were counted every 3 days post-dox. This experiment was repeated 2 times with 3 biological replicates in each experiment.

(D) Annexin V^+DAPI^- population in dox-inducible *Ren* or *Setd1a* shRNA-expressing cells was assessed at 4 days post-dox. This experiment was repeated 2 times with 3 biological replicates in each experiment.

(E) Dox-inducible *Ren* or *Setd1a* shRNA-expressing cells were stained with May-Grünwald/Giemsa at 4 days post-dox. Representative images from 3 independent experiments are shown. The bottom graph indicates the percentage of blasts (red), monocytes/macrophages (gray), and neutrophils (black).

(F) Western blot analysis of Ren (*Ren/Ren*), Setd1b (*Ren/Setd1b*), Setd1a (*Setd1a/Ren*), and *Setd1a/b* (*Setd1a/Setd1b*) shRNA-expressing cells was performed at 4 days post-dox. Quantified intensity is normalized by GAPDH signal, and fold changes are shown as a relative value to control. Representative images from 2 independent experiments are shown.

(G) Percentage of GFP+ shRNA-expressing leukemia cells in PB was analyzed at 3 weeks post-transplant. Representative data from 1 out of 2 independent experiments are shown.

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Notable functional differences between closely related SET1 homologs have been characterized in leukemia. For instance, although MLL3 and MLL4 are homologs of *Drosophila* Trithorax-related (Trr), MLL4 has a tumor-promoting role in *MLL*-r leukemia (Santos et al., 2014), while MLL3 is a tumor suppressor in −7/del(7q) AML (Chen et al., 2014). Differences such as these highlight the importance of defining unique functions of SET1 homologs. Homolog-specific binding proteins are likely one of the factors that influence the biologic activity of the different MLL complexes. For instance, MEN1 (also known as Menin) is a specific binding partner of MLL1/MLL2, and it is essential for chromatin binding and leukemogenic activity of MLL1 fusion proteins (Hughes et al., 2004; Yokoyama et al., 2004). Importantly, drugs targeting SET1 homolog binding partners have been identified as potential therapeutic tools against *MLL*-r/non-*MLL*-r leukemia and other solid cancers (Borkin et al., 2015; Cao et al., 2014; Funato et al., 2014; Grembecka et al., 2012; Kühn et al., 2016). Thus, to identify novel therapeutic drug targets in cancer, we need to better understand the unique protein-protein interactions and the functional roles of H3K4 HMT complexes in normal cells and in cancer cells.

Here, we report the essential roles of SETD1A in cell proliferation and define a novel protein-interacting domain in SETD1A. We show that the domain is required for the regulation of DNA damage response genes and that this novel domain binds cyclin K. Our data also demonstrate that SETD1A/cyclin K control DNA damage response gene expression during the S phase of the cell cycle. These results reveal a non-redundant and non-enzymatic role of SETD1A and suggest a potential therapeutic opportunity for leukemia and potentially other tumors.

RESULTS

SETD1A/COMPASS Complex Components Are Required for Growth of MLL-AF9 Leukemia Cells

To identify MLL/SET/COMAPSS complex subunits important for leukemia cell growth, we performed a small hairpin RNA (shRNA)-based screen with murine MLL-AF9 leukemia cells. We prepared 42 individual-shRNA vectors against MLL family members and COMPASS subunits and then performed a growth competition assay (Figure 1A). Toxicity from *Setd1a*, *Cxxc1*, *Hcfc1*, and *Wdr5* shRNAs suggested that the SETD1A/COMPASS complex might have critical roles in MLL-AF9 leukemia cells (Figure 1A). Efficacy and specificity of shRNAs were confirmed by rescue experiments with an exogenous SETD1A construct (Figures S1A–S1C). To evaluate the effect of *Setd1a* knockdown in greater detail, we generated murine MLL-AF9 leukemia single-cell clones that highly express a doxycycline-inducible *Setd1a* shRNA. After doxycycline administration, two *Setd1a* shRNAs showed significant suppression of *Setd1a* expression (Figure 1B), and inhibition of leukemia cell proliferation (Figure 1C). *Setd1a* shRNAs induced apoptosis, cell-cycle arrest, and neutrophilic differentiation (Figures 1D, 1E, and S1D). To examine and compare the functional role of *Setd1a* and *Setd1b* on H3K4 methylation, we also performed an inducible knockdown of either *Setd1a* or *Setd1b*, or both genes in murine MLL-AF9 leukemia cells (Figures 1F and S1E–S1I). Even in this setting, *Setd1a* knockdown predominantly induced growth arrest and cell differentiation (Figures S1H and S1I). Surprisingly, *Setd1a* knockdown did not show reduction of global H3K4me3 (Figure 1F). H3K4me3 chromatin immunoprecipitation sequencing (ChIP-seq) also showed no obvious change in H3K4me3 after knockdown of *Setd1a* (Figure S1J). To confirm these effects on other leukemia cells, we suppressed *SETD1A* in the MOLM-13 human leukemia cell line (Figures S1K). Consistent with murine cells, *SETD1A* shRNA expression in MOLM-13 cells induced growth arrest and differentiation (Figures S1L and S1M). To examine a role for SETD1A in leukemia progression *in vivo*, we performed a *Setd1a* knockdown experiment in a MLL-AF9 murine leukemia model. At 3 weeks post-transplantation, *Setd1a* shRNAs significantly suppressed the frequency of leukemia cells in the peripheral blood (PB) (Figure 1G). In addition, *Setd1a* shRNA constructs significantly prolonged the survival of recipient mice (Figures 1H). Consistent with data from the PB, *Setd1a* knockdown strongly suppressed the growth of leukemia cells in bone marrow (Figure S1N). In mice that did get sick in the *Setd1a* knockdown group, we did not observe GFP++ cells in their bone marrow (BM) (Figure S1O), demonstrating loss of *Setd1a* shRNA expression. When we sorted GFP++ cells expressing *Setd1a* shRNAs from the bone marrow and spleen at earlier time points, these cells showed comparable H3K4me3 levels (Figure S1P), lower *Setd1a* expression (Figure S1Q), higher percentage of apoptotic cells (Figure 1I), and decreased proliferation (Figure S1R). The morphology and Gr-1 expression of the GFP++ cells also indicated induction of differentiation by *Setd1a* suppression *in vivo* (Figures 1J and S1S). These data show that *Setd1a* has indispensable roles in leukemia cell proliferation and leukemia progression.

An Internal Region of SETD1A, but Not the SET Domain, Is Indispensable for Leukemia Growth

To evaluate the functional dependency on SETD1A, we performed rescue experiments with exogenous SETD1A mutant constructs (Figures 2A, 2B, and S2A). As expected, wild-type SETD1A expression restored the growth of RFP+ *Setd1a* shRNA-expressing cells (Figure S2B). Although the RRM domain deletion mutant failed to fully rescue cell growth at 12 days post-doxycycline, deletion of NSET and SET domains completely rescued the growth of *Setd1a* knockdown cells (Figures S2C). These results show that the H3K4 HMT activity of SETD1A is dispensable for MLL-AF9 leukemia growth.

(H) Survival of recipient mice harboring dox-inducible shRNA-expressing leukemia cells was plotted. Representative data from 1 out of 2 independent experiments are shown (n = 5/group).

(I) Annexin V^+DAPI^- population in *Ren* or *Setd1a* shRNA-expressing (GFP^{2+}, see also Figure S1N) leukemia cells in BM and spleen was analyzed at 3 weeks post-transplant. Representative data from 1 out of 2 independent experiments are shown (n = 4/group).

(J) *Ren* or *Setd1a* shRNA-expressing leukemia cells in BM were sorted and stained with May-Grünwald/Giemsa at 3 weeks post-transplant. Representative images of 3 biological replicates are shown.

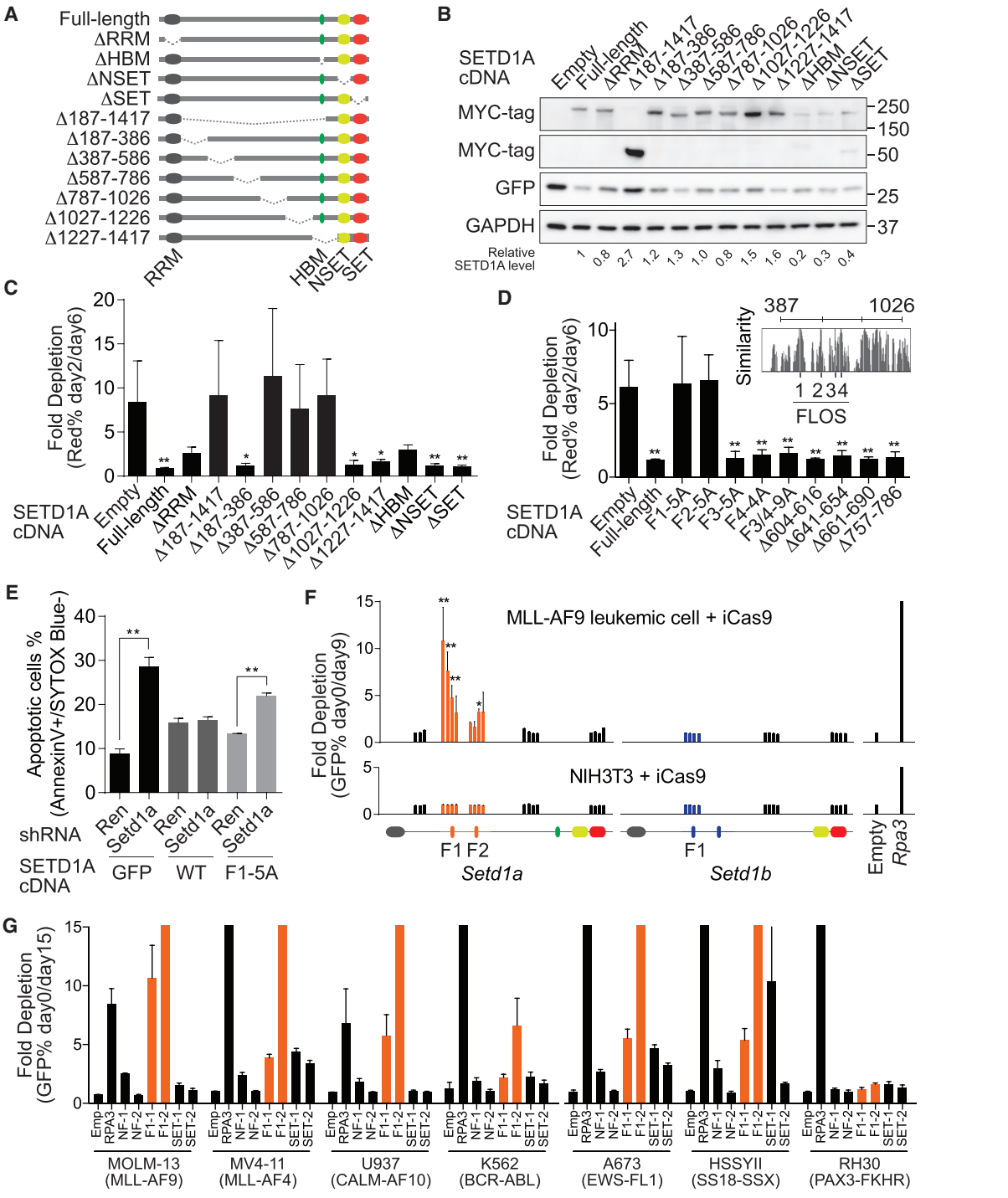


Figure 2. Internal Region on SETD1A Encodes a Critical Functional Domain for Leukemia Cell Survival

(A) The human SETD1A deletion mutant constructs are shown as schematic illustrations.

(B) Western blot analysis was performed using SETD1A deletion mutant-expressing 293T cells. These constructs were used for the following rescue experiments. Representative images from 2 independent experiments are shown.

(C) Human SETD1A deletion mutant-expressing mouse MLL-AF9 leukemia cells were transfected with dox-inducible mouse *Setd1a* 3′ UTR-targeting shRNAs. This experiment was performed with 2 independent shRNAs and repeated 2 times with 3 biological replicates in each experiment.

(D) Evolutionarily conserved motifs were identified inside of the functional region on SETD1A and named as FLOS1–4 (upper right panel). SETD1A mutant-expressing mouse MLL-AF9 leukemia cells were transfected with mouse Setd1a shRNAs. This experiment was performed with 2 independent shRNAs and repeated 2 times with 3 biological replicates in each experiment.

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The RRM domain of SETD1A is located at the N terminus, and the NSET and SET domains are located at the C terminus. Any role for the large internal region of SETD1A is unknown. We constructed eight additional SETD1A mutants, lacking parts of its internal region, and performed rescue experiments (Figures 2A and 2B). The SETD1A constructs that encode proteins lacking the entire internal region, as well as three independent deletion mutants (Δ387–586, Δ587–786, and Δ787–1026) showed strong functional defects (Figure 2C). These three mutant proteins were still mainly localized in the nucleus (Figure S2D). To identify potential functional domains spanning these regions, we compared the amino acid sequences of SETD1A and SETD1B across vertebrate species (Figures 2D and S2E). We found four highly conserved regions within 387–786 amino acids (aa) in SETD1A (Figure S2E). Here, we named these regions FLOS1–4 (functional location on SETD1A: F1–4) and constructed alanine-substituted SETD1A mutants. Alanine-substituted mutants of FLOS1 and FLOS2 ablated SETD1A function (Figure 2D). We constructed seven independent mutants spanning the aa region 587–786 in SETD1A. However, we found that every construct rescued leukemia cell growth after *Setd1a* knockdown (Figure 2D). These data suggest that the most critical regions of aa 387–1026 are FLOS1 and FLOS2. Furthermore, FLOS1 mutated SETD1A was unable to suppress apoptosis after *Setd1a* knockdown (Figure 2E). We have thus identified a novel functional region of SETD1A.

The FLOS Domain Supports Leukemia Cell Growth

To further validate the FLOS domain as critical for leukemia cell survival, we targeted the FLOS and SET domain of endogenous SETD1A protein in leukemia cells using a domain-focused CRISPR-mediated mutagenesis approach (Shi et al., 2015). Only FLOS domain targeted MLL-AF9 leukemia cells were significantly depleted. Cells with other *Setd1a*-targeted single guide RNAs (sgRNAs), as well as *Setd1b* targeting sgRNAs, were not depleted (Figures 2F and S2F). To assess whether targeting these specific regions of SETD1A suppressed growth of all proliferating cells, we also analyzed the roles of these domains in NIH 3T3 fibroblast cells. Remarkably, the FLOS domain targeting sgRNAs did not affect the ratio of GFP-positive cells in growth competition assays (Figure 2F). In contrast, an *Rpa3* sgRNA, which targets an essential gene, showed strong toxicity even in NIH 3T3 cells (Figure 2F). We performed both a SURVEYOR assay and deep-sequencing analysis of mutation abundance following doxycycline treatment (Figures S2G–S2J). As expected, both MLL-AF9 cells and NIH 3T3 cells showed high mutation frequencies, and the mutations from non-functional region targeted cells (NF-1/2) were stably detected in culture over time (Figures S2H–S2J). Although sgRNA-expressing NIH 3T3 cells still harbored the FLOS1 mutated allele 12 days after doxycycline treatment, MLL-AF9 leukemia cells did not show the FLOS1 mutated allele 9 days after doxycycline treatment (Figures S2G, S2I, and S2J). FLOS1/2 targeting sgRNAs were also tolerated in normal hematopoietic stem/progenitor cells, which formed colonies *in vitro* (Figure S2K). We also designed sgRNA constructs for the human SETD1A FLOS1 domain and expressed them in two MLL-r leukemia (MOLM-13, MV4-11), two non-MLL-r leukemia (U937, K562), and three sarcoma cell lines (A673, HSSYII, RH30) (Figure 2G). Other than RH30, the cell lines were sensitive to sgRNAs, which target the FLOS1 domain, but not the SET domain. These studies indicate that the FLOS domain of endogenous SETD1A is more critical for cancer cells than other cell types such as fibroblasts.

To confirm the endogenous SETD1A function in leukemia, we generated MLL-AF9 transformed *Setd1a*$^{fl/fl}$;*CreER*^{T2} cells (Figure 3A). After tamoxifen treatment, *Setd1a*$^{fl/fl}$ cells showed significant loss of colony-forming ability in culture (Figures 3B and S3A). Wild-type or SET domain deletion mutants completely rescued colony formation after *Setd1a* deletion, but neither FLOS1/2 mutants of SETD1A nor wild-type SETD1B rescued colony forming activity (Figure 3B). These results suggest that the FLOS domain may be responsible for the unique function of SETD1A in leukemia. To evaluate this, we performed domain swap experiments using expression constructs encoding a chimeric protein with SETD1B RRM/NSET/SET domains and SETD1A FLOS domain (Figures 3C and S3B). SETD1B variants showed comparable expression level in *Setd1a*$^{fl/fl}$;*CreER*^{T2} leukemia cells (Figures 3D). The SETD1 chimera encoding the 387–586 aa from SETD1A (B-FS) strongly supports cell survival and colony formation. The SETD1 chimera encoding the 387–1026 aa from SETD1A (B-FL) completely compensate the function of SETD1A (Figures 3E–3G, S3C, and S3D). SETD1 chimera-expressing *Setd1a*$^{fl/fl}$;*CreER*^{T2} leukemia cells showed complete deletion of *Setd1a* alleles (Figure 3H). These results indicate that the FLOS1 domain supports a non-redundant and non-enzymatic function of SETD1A in leukemia cell growth.

Disruption of SETD1A Suppresses the DNA Damage Response and Induces p53-Dependent Apoptosis in AML Cells

We performed RNA sequencing (RNA-seq) analysis using *Setd1a*$^{+/+}$ and *Setd1a*$^{Δ/Δ}$ MLL-AF9 leukemia cells (Figure 4A). Through gene ontology (GO) analysis and gene set enrichment analysis (GSEA), we found a significant decrease of DNA repair and Fanconi pathway-associated genes and a significant increase of p53 (encoded by *Trp53*) targets in *Setd1a*$^{Δ/Δ}$ MLL-AF9 leukemia cells (Figures 4B, S4A, and S4B). Genes critical

(E) Empty (GFP), SETD1A wild-type (WT), and FLOS1/alanine mutant (F1-5A)-expressing cells were transfected with mouse *Setd1a* 3′UTR shRNAs, and the annexin V^{+}DAPI^{-} populations were analyzed at 4 days post-dox. This experiment was performed with 2 independent shRNAs and repeated 2 times with 3 biological replicates in each experiment.

(F) A domain-focused CRISPR strategy was performed with *Setd1a* or *Setd1b* sgRNAs in inducible Cas9 (iCas9)-expressing MLL-AF9 leukemia cells or NIH 3T3 cells. The relative location of each sgRNA to the SETD1A or SETD1B protein is indicated along the x axis. Empty vector or *Rpa3* sgRNAs were used as negative or positive controls, respectively. Representative data from 1 out of 3 independent experiments are shown.

(G) A domain-focused CRISPR strategy was performed with *Setd1a* sgRNAs in human leukemia and sarcoma cell lines. Empty vector or *RPA3* sgRNAs were used as negative or positive controls, respectively. This experiment was repeated 2 times with 3 biological replicates in each experiment.

See also Figure S2.

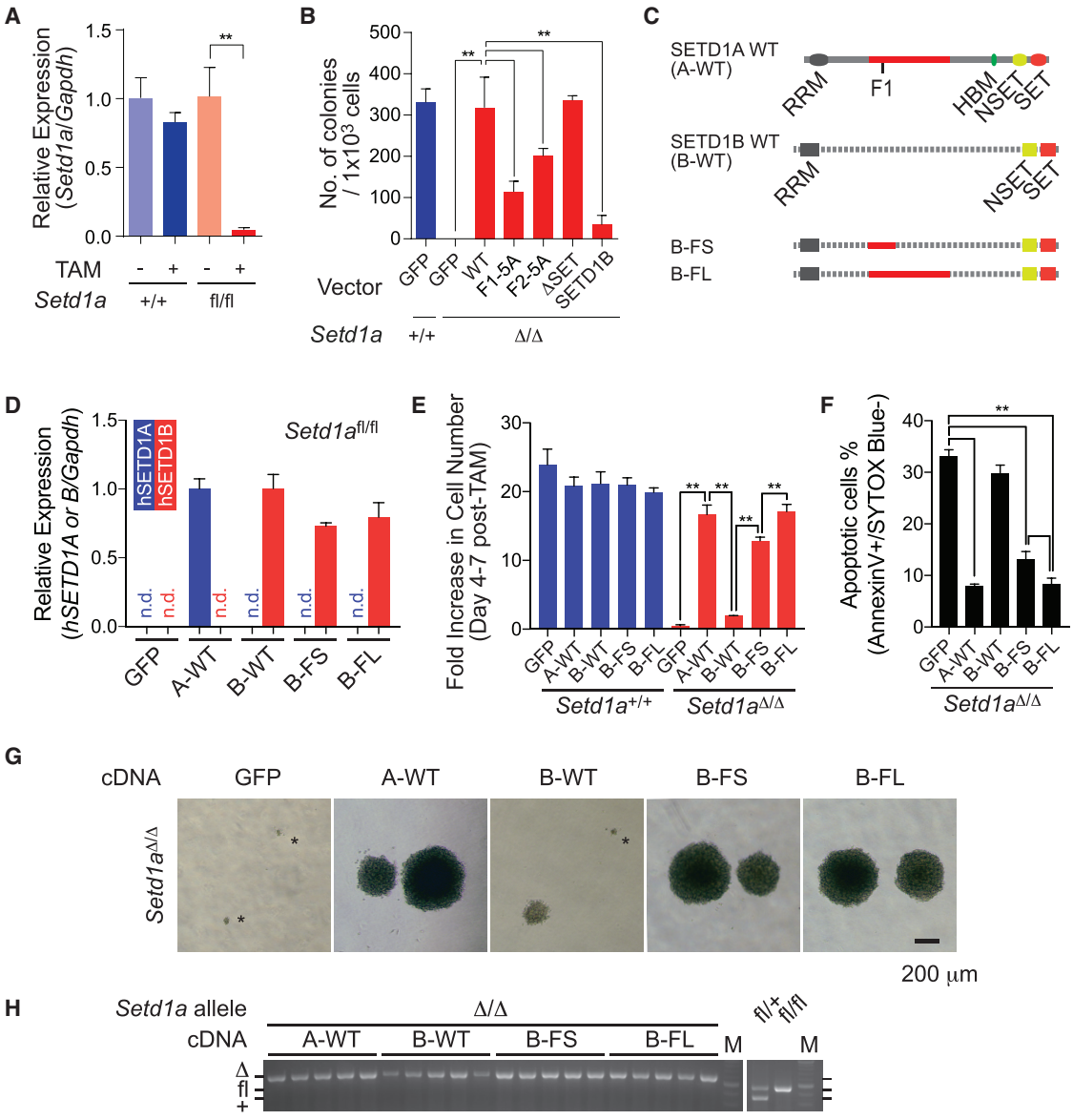


Figure 3. SETD1A FLOS Domain Provides a Non-redundant Role in Leukemia Cell Proliferation

(A) Relative expression level of *Setd1a* in MLL-AF9 leukemia cells was quantified by qRT-PCR at 4 days post-tamoxifen treatment. This experiment was repeated 3 times with 3 biological replicates in each experiment.

(B) MLL-AF9 leukemia cells were transfected with SETD1A mutant constructs. The cells were treated with tamoxifen and then assayed for colony-forming potential. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.

(C) The chimeric constructs of human SETD1A and human SETD1B are shown as schematic illustrations.

(D) *Setd1a*$^{fl/fl}$;*CreER* cells were transfected with SETD1 chimeric constructs and relative exogenous gene expression level was quantified by Taqman probes that can detect either human SETD1A or SETD1B. This experiment was repeated 2 times with 3 biological replicates in each experiment.

(E) MLL-AF9 leukemia cells were transfected with SETD1 chimeric constructs and monitored for cell number increase within 3 days from day 4 post-tamoxifen treatment. This experiment was repeated 3 times with 3 biological replicates in each experiment.

(F) Annexin V$^+$DAPI$^-$ population in *Setd1a*$^{fl/fl}$;*CreER* cells expressing SETD1 chimeric constructs was analyzed at day 5 post-tamoxifen treatment. This experiment was repeated 3 times with 3 biological replicates in each experiment.

(G) Representative morphology of colonies in *Setd1a*$^{fl/fl}$;*CreER* cells expressing SETD1 chimeric constructs are shown.

(H) Single-cell-derived colonies of *Setd1a*$^{fl/fl}$;*CreER* cells expressing SETD1 chimeric constructs were isolated after tamoxifen-treatment followed by colony-forming assay. *Setd1a* alleles of 5 clones are genotyped by PCR.

See also Figure S3.

for the DNA damage response including *Fancd2* and *Mlh1* were downregulated in *Setd1a*$^{\Delta/\Delta}$ cells, and these findings were confirmed by western blotting (Figures S4C) (Ceccaldi et al., 2016). *Fancd2* expression was downregulated 24 hr after *Setd1a* inactivation and preceded apoptosis induction (Figure 4C and S4D) supporting a direct effect of *Setd1a* suppression on *Fancd2* expression. Complete knockout of *Setd1a* also did not affect the abundance of H3K4me3 and H3K27Ac marks on the *Fancd2* or *Mlh1* locus (Figures 4D, S4E, and S4F; Table S2). Expression of wild-type SETD1A rescued *Fancd2* expression, whereas FLOS domain mutants did not rescue expression (Figure 4E). SETD1A/B chimeras containing the SETD1A FLOS1 domain also rescued *Fancd2* expression (Figure 4F). Consistent with a role in the DNA damage response, *Setd1a*$^{\Delta/\Delta}$ MLL-AF9 leukemia cells showed higher frequencies of chromosomal gaps/breaks (Figure 4G; Table S1). The SETD1A FLOS domain is therefore required for maintenance of genes that are important for the DNA damage response.

Since *Fancd2* null HSCs can be rescued by p53 suppression (Ceccaldi et al., 2012), we wanted to determine if *Setd1a* inactivation induced p53-dependent cell death. First, we found that p53 accumulated after *Setd1a* deletion (Figure 4H). To assess a potential functional role for p53 in *Setd1a*-induced apoptosis, we performed serial disruption of p53 (Figure S4G) and then the SETD1A FLOS1 domain by CRISPR. Remarkably, leukemia cells that harbored *p53* sgRNAs were resistant to the effects of two independent *Setd1a* sgRNAs targeting the FLOS1 domain, whereas control cells harboring sgRNAs targeting the *Rosa* locus were sensitive to the *Setd1a* sgRNAs (Figures 4I and S4H). p53 dependency was also confirmed with MLL-AF9/Cas9-expressing cells derived from p53$^{-/-}$ BM LSK (lineage$^-$ Sca-1$^+$ c-KIT$^+$) cells (Figure 4J). p53 knockdown did not affect the silencing of *Fancd2* in *Setd1a*-deficient cells (Figure 4K). These results demonstrate that SETD1A is required to regulate the expression of genes important for the DNA damage response and to protect leukemia cell chromosomal integrity, thus preventing p53-dependent apoptosis.

The FLOS Domain of SETD1A Binds Cyclin K

Having identified this critical domain in SETD1A, we set out to identify proteins that bound to this region. We expressed Flag-tagged SETD1A in 293T cells and performed co-immunoprecipitation (coIP) followed by mass spectrometry (MS) (Figure 5A). We identified 68 potential SETD1A binding proteins including several known SETD1A binding partners. GO analysis found that the proteins involved in the DNA damage response were enriched in the SETD1A binding proteins (Figures 5A and S5A). Of particular interest was cyclin K (encoded by *CCNK*), which is known to control transcription of multiple genes involved in the DNA damage response much like we have found for SETD1A (Blazek et al., 2011). To evaluate the cyclin K binding region in SETD1A, we performed coIP experiments in 293T cells using the SETD1A mutants in Figure 2A. Wild-type SETD1A was co-isolated with endogenous cyclin K, but SETD1A lacking aa 387–586 completely lost cyclin K-binding (Figure 5B). Remarkably, a fragment of SETD1A from aa 387–586 could be co-isolated with cyclin K and this association was disrupted by deletion or alanine substitution mutants of the FLOS1 domain (Figure 5C). Mutation of FLOS1, but not FLOS2, in full-length SETD1A was sufficient to deplete cyclin K binding (Figures 5D). FLOS1/2 mutation did not decrease the binding with COMPASS subunits, other known SETD1A complex subunits, or TCERG1, a RNAP2 regulator, which are all detected in MS (Figure 5D). Interestingly, strong cyclin K association was only observed with SETD1A, but not SETD1B, indicating that the unique interaction between SETD1A and cyclin K might provide SETD1A-specific functions in leukemia cells (Figure 5E). Deletion of the N- or C-terminal portions surrounding FLOS1 also disrupted the binding of cyclin K, suggesting that the structures surrounding the FLOS1 core sequence also support the interaction between SETD1A and cyclin K (Figure S5B). Cyclin K is observed at the H3K4me3-positive transcription start site (TSS), and *Setd1a* disruption reduced the signal for cyclin K, but not H3K4me3 (Figure 5F and S5C). To identify the SETD1A binding domain in cyclin K, we performed coIP experiments with domain deletion mutants of cyclin K, and we identified the cyclin box domains, which are required for the regulation of CDK12/13 kinase activity, as the binding region for SETD1A (Figure 5G). These results show that FLOS1 is critical for association with cyclin K and that the SETD1A-cyclin K pathway is indispensable for leukemia cell growth.

SETD1A-Cyclin K/CDK12 Axis Is Indispensable for AML Cells

Having identified the interaction between SETD1A and cyclin K, we wanted to assess the functional role of cyclin K in leukemia. We first performed a CRISPR domain scan using 188 sgRNAs spanning the cyclin K coding region and discovered an indispensable role for the cyclin box domains in leukemia cell proliferation (Figure 6A). Next, we mutated cyclin K using two sgRNAs that we identified in the CRISPR domain scan, and its CDK partners, CDK12/13, using 4 sgRNAs each (Figure 6B). The sgRNAs targeting kinase domains of CDK12 showed strong negative selection in leukemia cells similar to cyclin K, whereas the guides targeting CDK13 were not selected against. These results support the idea that the cyclin K/CDK12 pathway is an effector of SETD1A function in leukemia. RNA-seq analysis of cells expressing sgRNAs that target cyclin K showed that 190 genes overlap with the downregulated gene set in *Setd1a* knockout cells (Figure 6C). DNA damage response genes, including *Fancd2*, *Atr*, and *Mlh1*, were commonly downregulated (Figures 6D, 6E, and S6A–S6D). Disruption of *Cdk12* also caused downregulation of *Fancd2*, further supporting the concept that SETD1A regulates the function of the cyclin K/CDK12 complex (Figure 6F). The doxycycline-inducible cyclin K shRNAs led to acute reduction of *Fancd2* mRNA levels in leukemia cells (Figures S6E and S6F). Similar to those of FLOS domain mutations, the effects of cyclin K sgRNAs were partially rescued by p53 depletion (Figure S6G). Both SETD1A and cyclin K interact with CDK12, which suggested the functional importance of a SETD1A/cyclin K/CDK12 pathway (Figure 6G). The cyclin K/CDK12 complex has been shown to regulate RNAP2 CTD phosphorylation (Blazek et al., 2011), but SETD1A disruption did not show strong effects on the global level of RNAP2 CTD phosphorylation or on cyclin K protein levels in MLL-r leukemia cells (Figure 6H). These studies show that the cyclin K/CDK12 pathway is required for leukemia

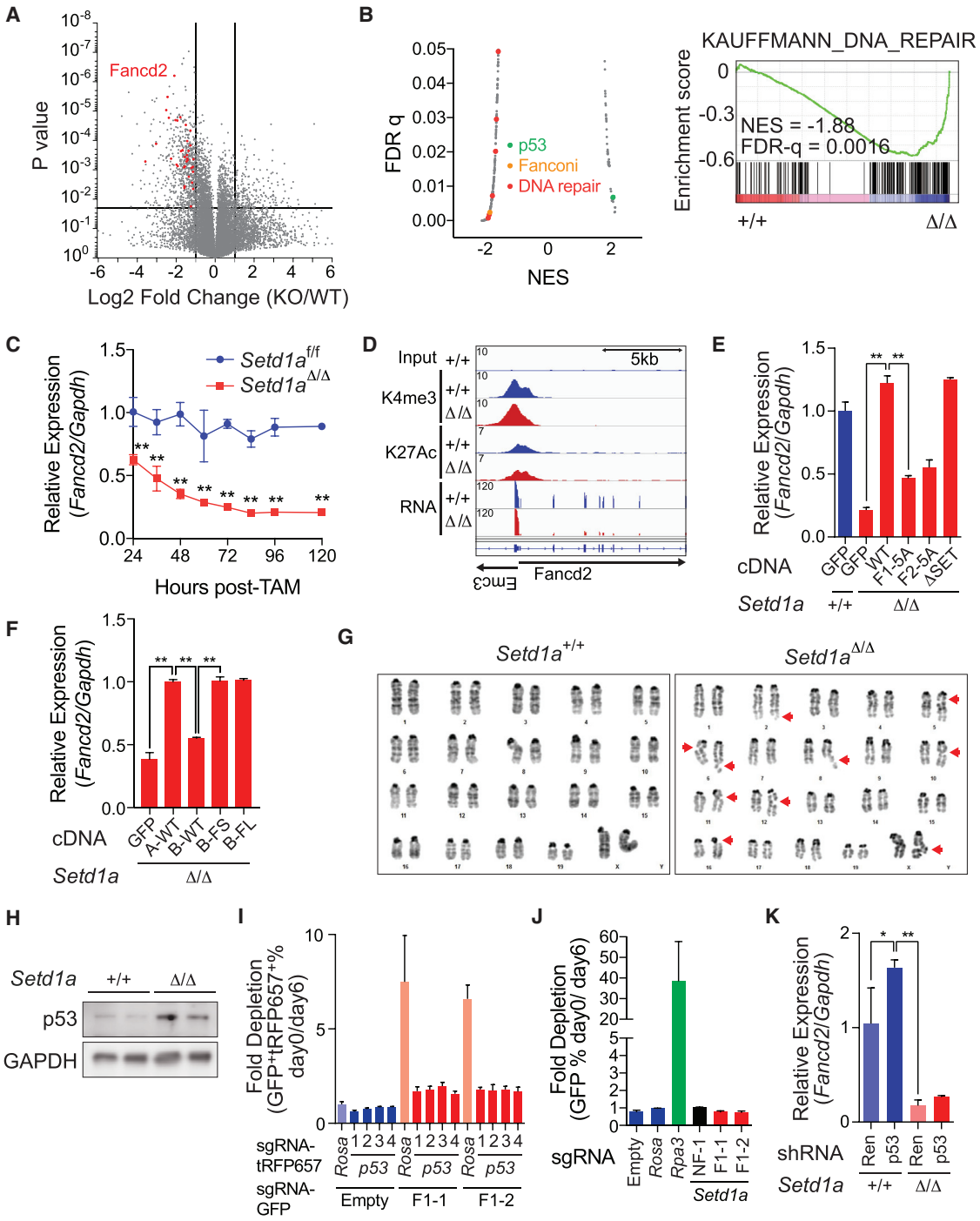


Figure 4. SETD1A FLOS Domain Regulates DNA Damage Response

(A) A volcano plot is used to depict RNA-seq data. Three independent samples on *Setd1a*$^{+/+}$ and *Setd1a*$^{Δ/Δ}$ leukemia cells were analyzed. Significantly downregulated DNA repair genes classified by GO analysis (see also Figure S4A) are shown as red dots.

(B) Gene set enrichment analysis was performed on RNA-seq data from MLL-AF9 leukemia cells. The scatterplot shows normalized enrichment score (NES) and false discovery rate (FDR) q-values comparing MSigDB curated gene set enrichment (left). Gene sets upregulated or downregulated in *Setd1a*$^{Δ/Δ}$ leukemia cells have positive or negative NES values, respectively. Representative enrichment plot from a DNA repair gene set is shown (right).

(C) qRT-PCR was performed to analyze *Fancd2* expression in *Setd1a*$^{fl/fl}$;*CreER* MLL-AF9 leukemia cells from 24 to 120 hr post-tamoxifen. This experiment was repeated 3 times with 3 biological replicates in each experiment.

(D) Histograms depict H3K4me3, H3K27Ac, and mRNA peaks on the *Fancd2* locus in *Setd1a* knockout leukemia cells.

(legend continued on next page)

cell growth similar to SETD1A, but that SETD1A does not affect global RNAP2 CTD phosphorylation.

SETD1A Regulates *FANCD2* Expression in a Cell-Cycle-Dependent Manner

Given that SETD1A and cyclin K are both required for appropriate expression of DNA damage response genes, but that global RNAP2 phosphorylation is not affected by SETD1A suppression, we wondered if the SETD1A function may be locus specific and therefore control cyclin K/CDK12 function at select loci. We assessed this possibility in NIH 3T3 cells. While NIH 3T3 cells showed significantly less sensitivity to *Setd1a* sgRNAs in competitive growth assays (Figure 2F), they still showed downregulation of FANCD2 and MLH1 expression, after *Setd1a* knockdown (Figure S7A). To examine the locus-dependent effects of the SETD1A-RNAP2 axis, we performed ChIP-seq analysis for SETD1A and RNAP2 in control and *Setd1a* knockdown NIH 3T3 cells (Figures 7A and 7B). SETD1A protein was detected on the promoter and TSS regions, and it decreased in cells where *Setd1a* was suppressed by shRNA knockdown (Figures 7A and 7B). In contrast, RNAP2 peaks were increased on the SETD1A-positive TSS in SETD1A knockdown cells (Figures 7A and 7B). Increasing of RNAP2 at TSS suggests promoter-proximal pausing after SETD1A knockdown. To monitor the cyclin K and RNAP2 CTD-Ser2 phosphorylation, we performed ChIP-qPCR at the *Fancd2* locus and confirmed the binding of SETD1A on the *Fancd2* TSS and observed significant decrease of cyclin K and RNAP2 CTD-Ser2 phosphorylation (Figure 7C, S7B, and S7C). These results show that SETD1A is crucial for the recruitment of cyclin K, which promotes CDK12 occupancy/activation, RNAP2 phosphorylation and gene expression.

Previous studies have shown that molecules such as FANCD2, a SETD1A-cyclin K target gene, are necessary in S phase of the cell cycle (Ceccaldi et al., 2016). We hypothesized that the SETD1A-cyclin K complex might regulate gene expression at the G1-S transition. We employed a system to monitor *FANCD2* transcription during the cell cycle in a manner that did not require molecules that induced DNA damage or cellular toxicity. We labeled the 293T cell line with FUCCI (fluorescent, ubiquitination-based cell-cycle indicator) and enriched the cells at each cell-cycle stage by cell sorting (Figure 7D) (Sakaue-Sawano et al., 2008). Interestingly, RNA as well as protein levels for FANCD2 were increased in the S-phase-enriched fraction (C+G+) (Figures 7E and 7F). Cyclin K-SETD1A interaction did not change through the G1-S transition (Figure S7D), but either SETD1A or cyclin K knockdown significantly suppressed the cell-cycle-dependent induction of *FANCD2* expression (Figures 7G, 7H, S7E, and S7F). In addition, pharmacological inhibition of CDK12/13 with THZ531, a covalent inhibitor (Zhang et al., 2016), also suppressed the transcriptional activation of *FANCD2* in S phase (Figure 7I).

SETD1A/Cyclin K/CDK12 Pathway Regulates Cell Growth of Human Leukemia Cells

To evaluate the effect of cyclin K/CDK12 suppression in human leukemia, we treated human leukemia cell lines as well as MLL-AF9 leukemia cells derived from human CD34+ cord blood with THZ531. THZ531 inhibited growth of MLL-r/non-MLL-r leukemia cell lines and induced apoptosis in the MOLM-13 cell line (Figures S7G and S7H). We also confirmed the importance of the SETD1A-CCNK axis in MLL-AF9-transduced human cord blood cells through shRNA-based suppression of either SETD1A or CCNK (Figure S7I). These results demonstrate that the SETD1A/cyclin K/CDK12 pathway regulates cell-cycle-dependent expression of DNA damage repair genes and that inhibition of these mechanisms leads to significant inhibition of growth of human leukemia cells.

DISCUSSION

There is increasing interest in the function of the MLL family of H3K4 HMTs as they have been shown to play diverse roles in organismal and cancer development (Rao and Dou, 2015). Furthermore, accumulating evidence suggests that multiple members of this family may be therapeutic targets in cancer (Cao et al., 2014; Chen et al., 2014; Kühn et al., 2016; Santos et al., 2014; Xu et al., 2016). The cellular roles of MLL family members is presumed to be due to H3K4 methylation, but in only a few studies has the HMT activity been demonstrated to be of importance. Indeed, recent studies indicate that the SET domain of MLL1 is dispensable for leukemia that requires MLL1 for continued proliferation (Kühn et al., 2016; Mishra et al., 2014). The data presented here show that leukemia cells tolerate inactivation of the SET domain of SETD1A even though they are absolutely dependent on SETD1A for continued proliferation and survival. Given that SETD1A and SETD1B are thought be responsible for much of the cellular H3K4 methylation, this

(E) MLL-AF9 leukemia cells were transfected with SETD1A mutant cDNA, and qRT-PCR analysis of *Fancd2* was performed at 4 days post-tamoxifen. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.
(F) MLL-AF9 leukemia cells were transfected with SETD1A/B mutant cDNA, and qRT-PCR analysis of *Fancd2* was performed at 4 days post-tamoxifen. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.
(G) Images are shown of representative karyotypes in control and *Setd1a*-deficient leukemia cells. Red arrows indicate breaks/gaps on the chromosome. See Table S1.
(H) MLL-AF9 leukemia cells were treated with tamoxifen, and p53 protein expression was analyzed by western blot analysis at 4 days post-tamoxifen.
(I) *p53* sgRNAs with a tRFP657 reporter and *Setd1a* FLOS1 sgRNAs with a GFP reporter were sequentially transfected into iCas9-expressing MLL-AF9 leukemia cells. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.
(J) *Setd1a* FLOS1 sgRNAs with a GFP reporter were transduced into *p53*−/− derived iCas9-expressing MLL-AF9 leukemia cells. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.
(K) MLL-AF9 leukemia cells were transfected with *p53* shRNA and treated with tamoxifen for 24 hr. Relative expression level of *Fancd2* was quantified by qRT-PCR at 4 days post-tamoxifen. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.
See also Table S2.

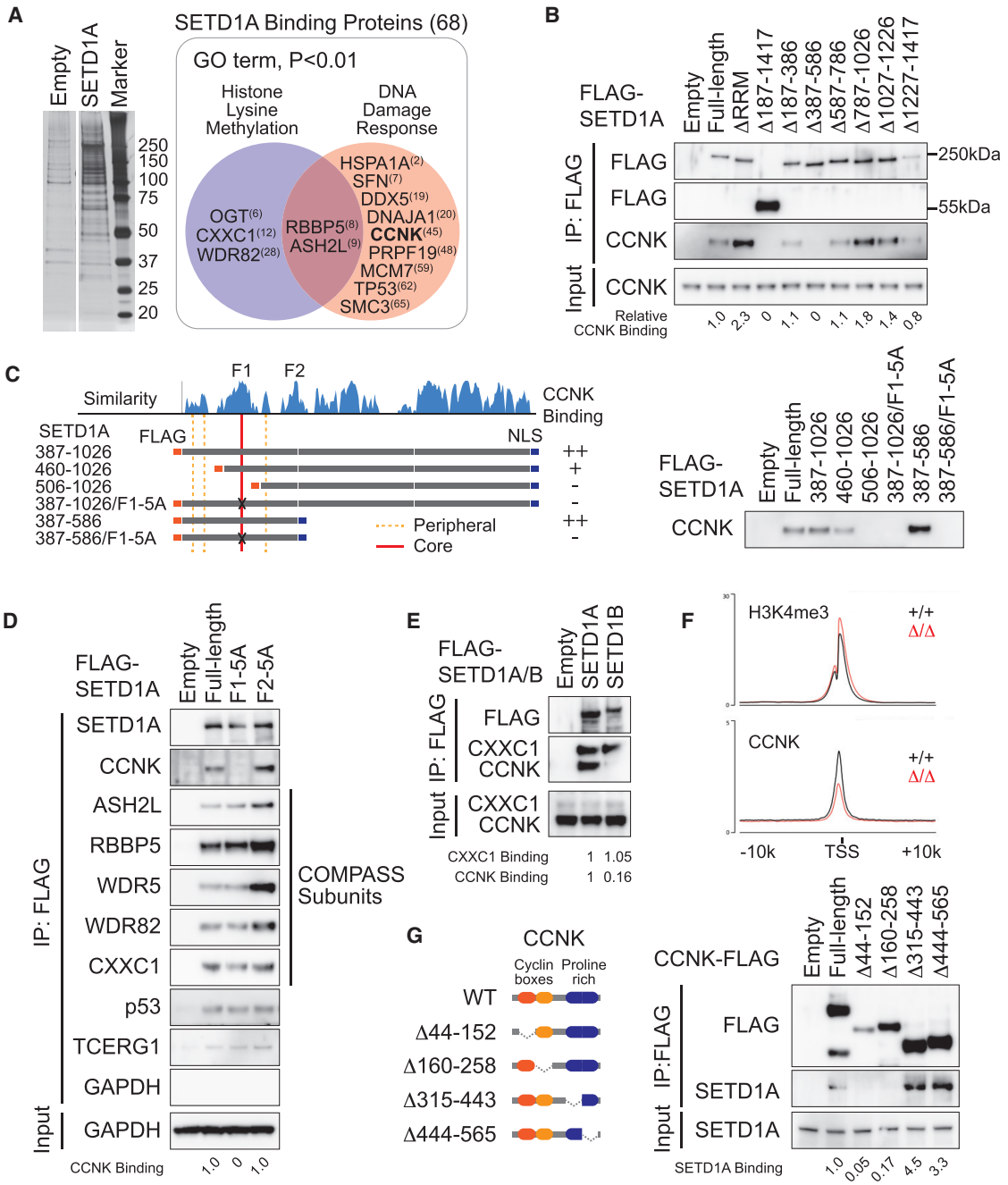


Figure 5. SETD1A FLOS Domain Binds Cyclin K

(A) SETD1A binding proteins were identified by coIP MS. Proteins were visualized by silver stain (left) and excised from the gel. 68 proteins were identified as SETD1A binding proteins (right box outlined in gray), and 11 proteins were classified as DNA damage response-associated proteins in GO analysis (red circle). The ranks of signal intensity are shown in brackets.

(B) 293T cells were transfected with SETD1A deletion mutant constructs, and the protein extracts were used for the coIP experiment.

(C) 293T cells were transfected with internal region-specific fragments of SETD1A constructs. FLAG-tagged fragments illustrated on the left were used for coIP.

(D) 293T cells were transfected with SETD1A FLOS1 or FLOS2 alanine mutant constructs.

(E) 293T cells were transfected with SETD1A and SETD1B constructs. CXXC1, a common binding partner of SETD1A/B, was detected as a positive control.

(F) Average ChIP-seq signal of H3K4me3 (upper) and CCNK (lower) at H3K4me3-positive TSS from *Setd1a*$^{\Delta/\Delta}$ and *Setd1a*$^{+/+}$ leukemia cells.

(G) 293T cells were transfected with cyclin K mutants. FLAG-tagged cyclin K mutants illustrated on the left were used for coIP.

See also Figure S5.

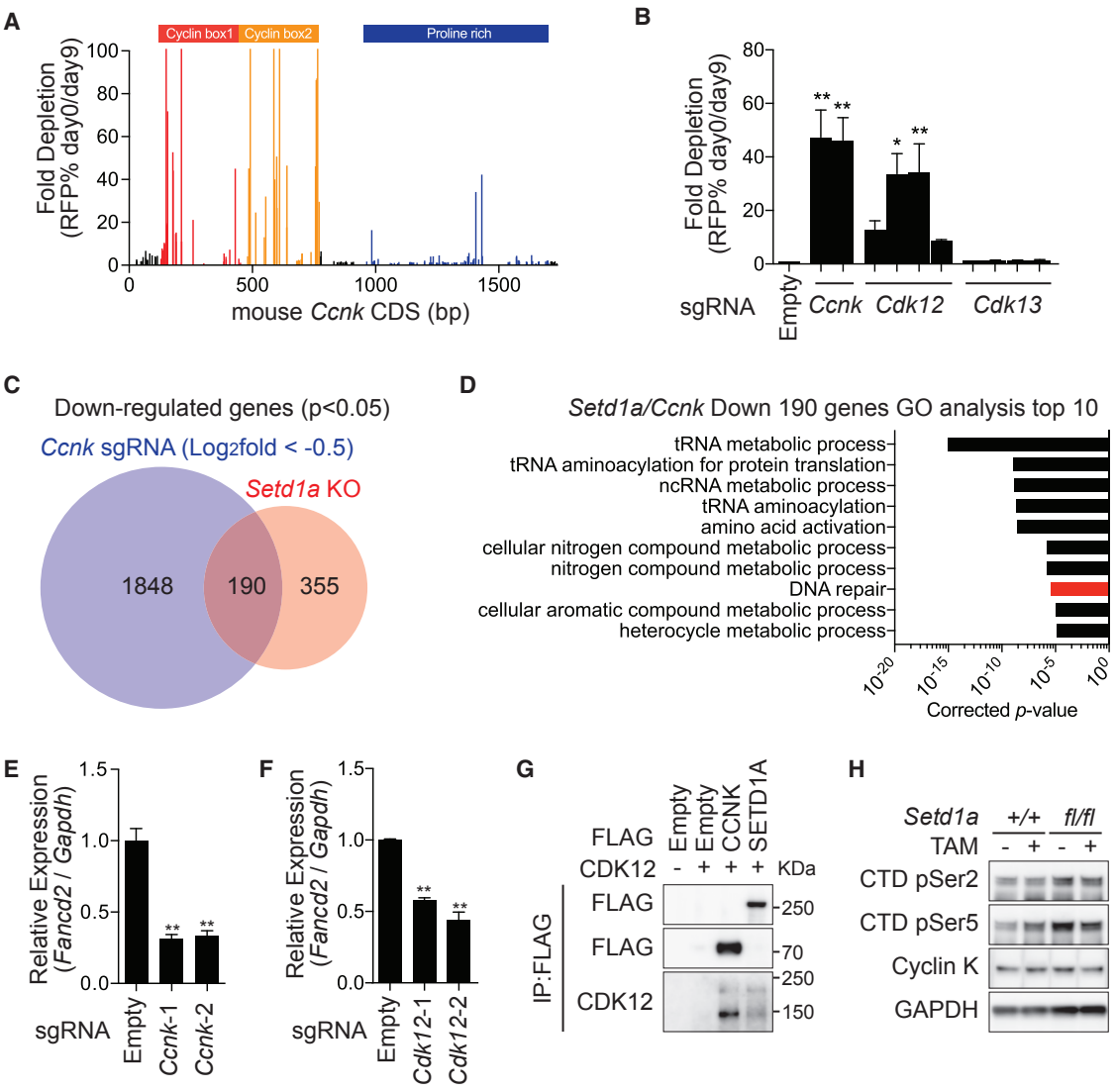


Figure 6. SETD1A Regulates the Cyclin K/CDK12 Axis in MLL-r Leukemia

(A) A domain-focused CRISPR strategy was used to analyze the functional regions of the *Ccnk* gene in mouse MLL-AF9 leukemia cells. Representative data from 1 out of 3 independent experiments are shown.

(B) *Cdk12/13* kinase domain-targeting sgRNAs with a RFP reporter were used for competitive growth assay in iCas9-expressing MLL-AF9 leukemia cells. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.

(C) RNA-seq analysis was performed with iCas9-expressing MLL-AF9 leukemia cells harboring *Ccnk* sgRNAs. Four independent *Ccnk* sgRNAs were used. The overlap in downregulated genes between *Ccnk* sgRNA-expressing cells and *Setd1a* knockout cells are shown.

(D) Gene ontology analysis of 190 commonly downregulated genes is shown in (C). The bar for DNA damage stimulus is shown in red.

(E and F) qRT-PCR analysis of *Fancd2* in *Ccnk* sgRNA (E) or *Cdk12* sgRNA (F)-expressing leukemia cells at 4 days post-dox. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.

(G) 293T cells were transfected with FLAG-tagged cyclin K or FLAG-tagged SETD1A and CDK12 expression vectors. FLAG-tagged antibody was used for the coIP experiment. Representative images from 2 independent experiments are shown.

(H) Western blot analysis was performed to check the global level of cyclin K and RNAP2 CTD (Ser2 or Ser5) phosphorylation in SETD1A-deficient MLL-AF9 leukemia cells. Representative images from 2 independent experiments are shown.

See also Figure S6.

suggests functional redundancy for H3K4 methylation between these highly homologous proteins. However, it is clear that there is a critical non-redundant function for SETD1A in leukemia cells that prompted further study. These data support the concept that MLL family members play critical roles in cell biology that are independent of H3K4 methylation and cautions against necessarily focusing on the SET domain as critical for a given phenotype.

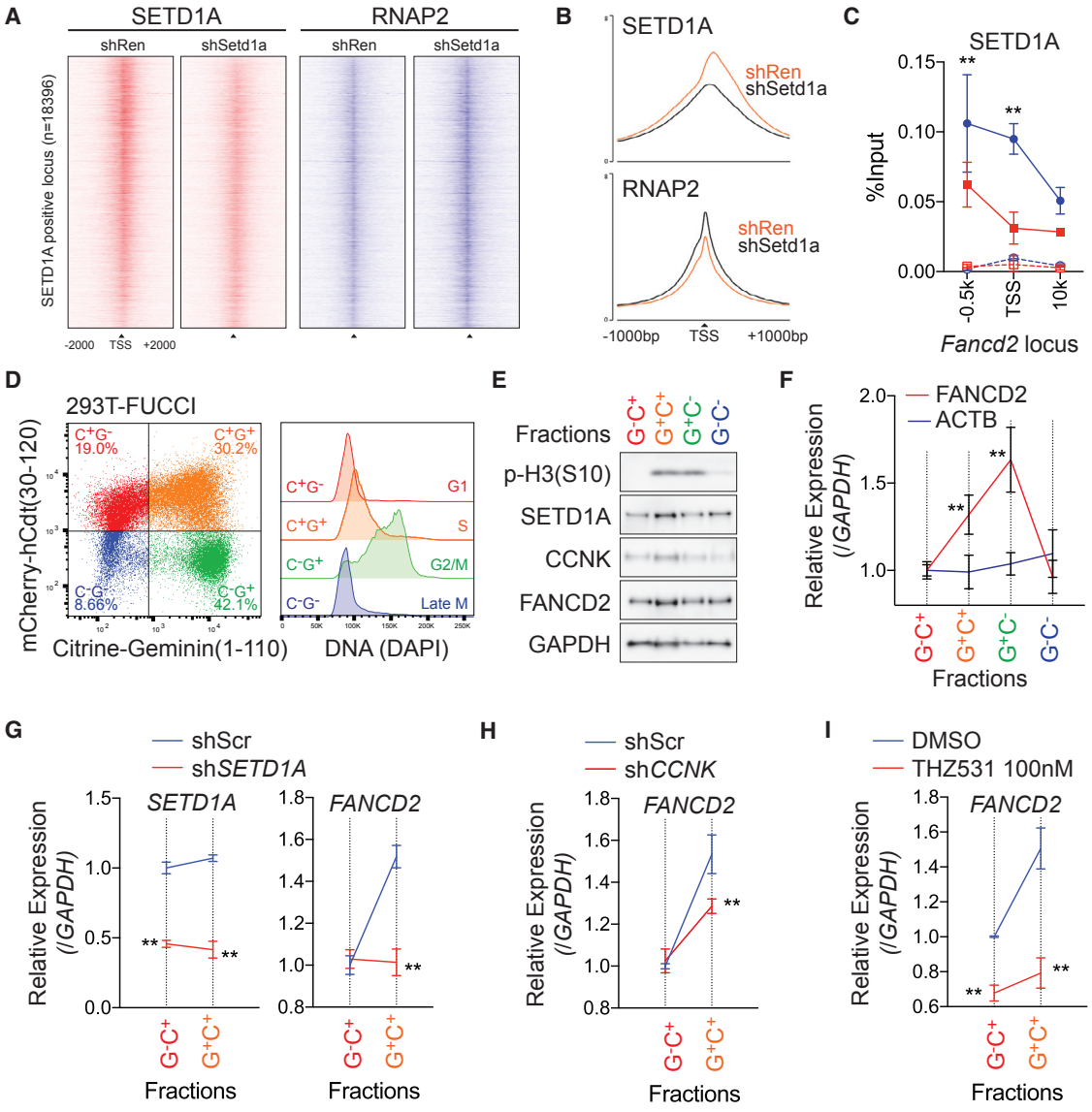


Figure 7. SETD1A-Cyclin K Regulates *FANCD2* Transcription in G1/S Phase

(A) Density of ChIP-seq reads for SETD1A (red) and RNAP2 (blue) of SETD1A-positive TSS (n = 18,396) from *Ren* shRNA or *Setd1a* shRNA-expressing NIH 3T3 cells are shown. Regions are ranked according to log2 fold change of SETD1A signal intensity between *Ren* and *Setd1a* shRNA.

(B) Average ChIP-seq signal of SETD1A (upper) and RNAP2 (lower) of SETD1A-positive TSS from *Ren* shRNA or *Setd1a* shRNA-expressing NIH 3T3 cells are shown.

(C) ChIP-qPCR was performed for SETD1A on the *Fancd2* locus in *Ren* shRNA (blue) or *Setd1a* shRNA (red)-expressing NIH 3T3 cells at 4 days post-dox. Rabbit IgG was used as negative control in ChIP (dot lines). Representative data from 3 independent experiments with 3 biological replicates are shown.

(D) Cell-cycle stage was visualized by flow cytometry in a FUCCI-positive 293T single-cell clone (left panel). DNA contents in each fraction are shown in the right panel.

(E) 293T-FUCCI cells were fractionated by cell sorting. Western blot analysis was performed using the indicated antibodies. Representative data from 3 independent experiments are shown.

(F) *FANCD2* transcriptional level during the cell cycle was monitored by qRT-PCR. This experiment was repeated 3 times with 3 biological replicates in each experiment.

(G) Scramble shRNA (Scr, blue) and SETD1A shRNA (red)-expressing 293T-FUCCI cells were analyzed at 4 days post-transfection. Expressions of *SETD1A* (left) and *FANCD2* (right) are shown. See also Figure S7E.

(H) *FANCD2* transcription in Scr shRNA (blue) and CCNK shRNA (red)-expressing 293T-FUCCI cells were analyzed at 4 days post-transfection. See also Figure S7F.

(I) *FANCD2* transcription in DMSO (blue) and 100 nM THZ531 (red)-treated 293T-FUCCI cells were analyzed at 3 days post-treatment. Representative data from 1 out of 2 independent experiments with 3 biological replicates are shown.

Through detailed mutagenesis and CRISPR-Cas9 domain scanning, we found a previously unidentified domain in SETD1A that is critical for leukemia proliferation and survival. We call this the functional location on SETD1A (FLOS) domain. The FLOS domain is critical for SETD1A-dependent expression of genes involved in the DNA damage response and interacts with cyclin K. This is in keeping with the recent demonstration that the cyclin K/CDK12 complex is a regulator of DNA damage response genes (Blazek et al., 2011; Greifenberg et al., 2016). It also shows the functional importance of the SETD1A/cyclin K interaction in leukemia cells. Importantly, SETD1A depletion reduces cyclin K occupancy on the *Fancd2* gene locus, without affecting the global cyclin K protein levels, thus demonstrating that the SETD1A FLOS domain is essential for locus-specific recruitment of cyclin K. These data are important not only because they highlight a relationship between SETD1A, cyclin K, and the DNA damage response but also because recent efforts toward the drug development of CDK12/13 inhibitors suggest this pathway may be an important target for cancer drug development (Zhang et al., 2016).

We show that SETD1A is required for appropriate expression of DNA damage response genes. Since *Fancd2* was the most sensitive gene to SETD1A disruption, we assessed *Fancd2* dependence in leukemia and found an essential role in the maintenance of MLL-AF9 leukemia cells. Activation of p53 is observed upon *Fancd2* deficiency and p53 knockout restores the *Fancd2*$^{-/-}$ BM LSK cell population and functions *in vivo* (Ceccaldi et al., 2012; Kim et al., 2003). Consistent with these results, cellular differentiation and cell death induced by targeting the FLOS domain was rescued by p53 knockout. Beyond *Fancd2*, *Atr* expression was also dependent on the SETD1A. *Atr* disruption also induces myeloid differentiation and loss of genome integrity in MLL-r leukemia cells much like SETD1A inactivation (Ruzankina et al., 2007; Santos et al., 2014). Our results strongly support the idea that SETD1A coordinately regulates DNA damage response pathways that are critical for leukemia proliferation via its FLOS domain.

Cyclin K was originally identified as a cyclin that could rescue the lethality of G1 cyclin-deficient yeast, but the cell-cycle-dependent functions of cyclin K have not been deeply characterized (Edwards et al., 1998). Here, we found the highest levels of cyclin K protein in S phase using FUCCI, a non-invasive cell-cycle indicator (Sakaue-Sawano et al., 2008). Moreover, our data demonstrate that SETD1A is functionally important for S-phase-dependent *FANCD2* expression. Our findings provide new insights into how epigenetic regulators functionally modulate gene transcription in a cell-cycle-dependent manner. Importantly, FLOS-domain-specific targeting was toxic in leukemia cells, but not fibroblast cells, suggesting that there may be an enhanced dependency in leukemia cells.

Taken together, our work shows that the SETD1A FLOS domain regulates expression of genes involved in the DNA damage response via interaction with cyclin K. Furthermore, MLL family HMTs can work as regulators of gene expression that are independent of their enzymatic functions. This newly defined function for SETD1A could be a novel therapeutic target for cancer drug development and therefore detailed structural analysis of the FLOS domain and its interaction with cyclin K will be illuminating and potentially important for future drug development. Finally, SETD1A is a multifunctional modulator of gene expression with mechanisms that are overlapping and non-overlapping with SETD1B. This is likely true for other MLL family members. This is an attribute that should be considered as drugs are developed to target this family of proteins.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental Information includes seven figures and four tables and can be found with this article online at https://doi.org/10.1016/j.cell.2018.01.032.

ACKNOWLEDGMENTS

We would like to thank Jessica Brady and Zhaohui Feng for administrative help with this project. We are grateful to Hideki Terai and Kunihiro Hinohara for technical advice. We are also grateful to Nathaneal Gray for providing THZ531. T.H. was supported by a grant from Kanae Foundation for the Promotion of Medical Science and the JSPS Program for Advancing Strategic International Networks to Accelerate the Circulation of Talented Researchers, JAPAN. S.W.L. was supported by an NCI grant (CA190261). A.K. was supported by NCI grants (CA204396 and P30 CA008748). S.A.A. was supported by the Leukemia and Lymphoma Society and NCI grants (CA066996 and CA140575).

AUTHOR CONTRIBUTIONS

Conceptualization, T.H. and S.A.A.; Investigation, T.H., P.C., and Z.F.; Resources, C.-H.H., C.-W.C., C.D.D., S.W.L., and A.K.; Formal Analysis, T.H. and R.K.; Writing-Original Draft, T.H.; Writing-Review & Editing, C.D.D. and S.A.A.; Supervision, S.A.A.; Funding Acquisition, S.A.A.

DECLARATION OF INTERESTS

S.A.A. consults for Epizyme, Imago Biosciences, and C4 Therapeutics. All other authors declare no competing interests.

Received: April 20, 2017
Revised: October 30, 2017
Accepted: January 24, 2018
Published: February 22, 2018

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-CXXC1	Abcam	Cat#ab56035, RRID: AB_879712
Rabbit polyclonal anti-H3K4me1	Abcam	Cat#ab8895, RRID: AB_306847
Rabbit polyclonal anti-H3K4me2	Millipore	Cat#07-030, RRID: AB_10099880
Rabbit polyclonal anti-H3K4me3	Abcam	Cat#ab8580, RRID: AB_306649
Rabbit polyclonal anti-H3K27Ac	Abcam	Cat#ab4729, RRID: AB_2118291
Rabbit polyclonal anti-Histone H3	Abcam	Cat#ab1791, RRID: AB_302613
Rabbit monoclonal anti-GAPDH	Cell Signaling	Cat#2118S, RRID: AB_561053
Rabbit monoclonal anti-WDR5	Abcam	Cat#ab178410
Rabbit monoclonal anti-ASH2L	Cell Signaling	Cat#5019S, RRID: AB_1950350
Rabbit polyclonal anti-RbBP5	Bethyl Laboratories	Cat#A300-109A, RRID: AB_210551
Rabbit polyclonal anti-WDR82	Abgent	Cat#AP4812a, RRID: AB_10663080
Rabbit polyclonal anti-CA150(TCERG1)	Bethyl Laboratories	Cat#A300-360A, RRID: AB_185568
Rabbit monoclonal anti-FANCD2	Abcam	Cat#ab108928, RRID: AB_10862535
Rabbit monoclonal anti-MLH1	Abcam	Cat#ab92312, RRID: AB_2049968
Mouse monoclonal anti-p53	Abcam	Cat#ab28, RRID: AB_303312
Rabbit polyclonal anti-FLAG	GenScript	Cat#A00170-40, RRID: AB_914510
Mouse monoclonal anti-FLAG M2 Affinity Gel	Sigma	Cat#A2220, RRID: AB_10063035
Rabbit polyclonal anti-Cyclin K	Bethyl Laboratories	Cat#A301-939A, RRID: AB_1547934
Rabbit polyclonal anti-SETD1A	Abcam	Cat#ab70378, RRID: AB_1951955
Rabbit polyclonal anti-RNA polymerase II CTD repeat phospho S2	Abcam	Cat#ab5095, RRID: AB_304749
Rabbit polyclonal anti-RNA polymerase II CTD repeat phosphor S5	Abcam	Cat#ab5131, RRID: AB_449369
Mouse monoclonal anti-RNA polymerase II	diagenode	Cat#C15100055
Rabbit polyclonal anti-Lamin B1	Abcam	Cat#ab16048, RRID: AB_443298
Chemicals, Peptides, and Recombinant Proteins		
Nutlin-3	Sigma	Cat#N6287
THZ531, CDK12/CDK13 inhibitor	Laboratory of Nathaneal Gray	N/A
Deposited Data		
Raw and analyzed data of RNA-seq and ChIP-seq	This paper	GEO: GSE108615
Raw data of MS analysis	This paper	MassIVE: MSV000080012
Experimental Models: Cell Lines		
Human: 293T cells	ATCC	CRL-3216
Human: Plat-E cells	Cell Biolabs	RV-101
Human: MV4-11 cells	ATCC	CRL-9591
Human: U937 cells	ATCC	CRL-1593.2
Human: K562 cells	ATCC	CCL-243
Human: KG-1 cells	ATCC	CCL-246
Human: HL-60 cells	ATCC	CCL-240
Human: THP-1 cells	ATCC	TIB-202
Human: MOLM-13 cells	DSMZ	ACC-554
Human: NB-4 cells	DSMZ	ACC-207
Human: ML-2 cells	DSMZ	ACC-15
Human: Jurkat cells	DSMZ	ACC-282

(*Continued on next page*)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human: A673 cells	ATCC	CRL-1598
Human: SJCRH30 (RH30) cells	ATCC	CRL-2061
Human: HSSYII cells	Laboratory of Stefan Fröhling	N/A
Experimental Models: Organisms/Strains		
Mouse: *Setd1a*$^{tm1a(EUCOMM)Wtsi}$	EMMA	EM: 06857
Mouse: *Act-Flpe*; *B6.Cg-Tg(ACTFLPe)9205Dym/J*	The Jackson Laboratory	JAX: 005703
Mouse: Rosa26-Cas9 knockin; Gt(ROSA)26Sortm1.1 (CAG-cas9*,-EGFP)Fezh/J	The Jackson Laboratory	JAX: 024858
Mouse: B6.129S2-Trp53^{tm1Tyj}/J	The Jackson Laboratory	JAX: 002101
Oligonucleotides		
shRNA (see Table S3)	This paper	N/A
sgRNA (see Table S3)	This paper	N/A
Genotyping DNA Primers (see Table S4)	This paper	N/A
TaqMan Probes (see Table S4)	Life Technologies	Cat#4331182, Cat#4351372
Recombinant DNA		
Plasmid: TRIN miR-E	Laboratory of Scott Lowe	N/A
Plasmid: RT3GEPIR miR-E	MSKCC RNAi Core	N/A
Plasmid: pMSCV-MLL-AF9-ires-Neo	Laboratory of Scott Armstrong	N/A
Plasmid: pMSCV-MLL-AF9-ires-GFP	Laboratory of Scott Armstrong	N/A
Plasmid: pMSCV-rtTA-pgk-hygro	Laboratory of Scott Armstrong	N/A
Plasmid: pLKO5.sgRNA.EFS.GFP	Heckl et al., 2014	Addgene #57822
Plasmid: pLKO5.sgRNA.EFS.tRFP657	Heckl et al., 2014	Addgene #57824
Plasmid: pCW-Cas9	Wang et al., 2014	Addgene #50661
Plasmid: pMD2.G	Laboratory of Didier Trono	Addgene #12259
Plasmid: psPAX2	Laboratory of Didier Trono	Addgene #12260
Plasmid: ES-FUCCI	Sladitschek and Neveu, 2015	Addgene #62451
Plasmid: pLKO.1 puro	Stewart et al., 2003	Addgene #8453
Plasmid: pCMV6-hSETD1A(NM_014712)-Myc-DDK	Origene	Cat#RC214996
Plasmid: pCVM6-hSETD1B(NM_015048)-Myc-DDK	Origene	Cat#RC227091
Plasmid: pcDNA3.1-CCNK(NM_001099402)-(K)DYK	GenScript	Cat#OHu30814D
Plasmid: pCI-Neo-His-Pitaire(CDK12)-Myc	Ko et al., 2001	Addgene #26066
Software and Algorithms		
Prism 7	GraphPad softare	N/A
CRISPResso ver 0.8.4	Pinello et al., 2016	https://github.com/lucapinello/CRISPResso
EaSeq	Lerdrup et al., 2016	http://easeq.net
Analysis of NGS data	Basepair software	https://www.basepairtech.com

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Scott A. Armstrong (scott_armstrong@dfci.harvard.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Mice

Setd1a$^{tm1a(EUCOMM)Wtsi}$ mice (EMMA) were crossed with *Act-Flpe* mice (The Jackson Laboratory, 005703) to generate a floxed allele. Rosa26-Cas9 knockin mice (024858) and p53 knockout mice (002101) were obtained from The Jackson Laboratory. C57BL/6 female mice (CD45.2, Charles River) were used as transplantation recipients. Mice used in this study have been maintained in the Research

Animal Resource Center of MSKCC and the DFCI Animal Resource Center, following a vertebrate animal protocol approved by MSKCC's and DFCI's Institutional Animal Care and Use Committee (IACUC), respectively. Both female and male mice were used in our leukemia studies and no obvious sex-dependent differences were observed.

Human Cord Blood Samples

Normal cord blood units, designated for research use, were obtained from the New York Blood Center (NYBC, New York, NY) under IRB approval from the National Cord Blood Program (NCBP). Both male and female samples were randomly assigned and used in this study. The experiments were approved by the institutional biosafety committees at MSKCC and DFCI.

Cell Lines and Cell Culture

293T, Plat-E, A673 and HSSYII cell lines were maintained in DMEM medium containing 10% fetal bovine serum (FBS) and 1% penicillin-streptomycin. MV4-11, K562, KG-1, HL-60 cell lines were maintained in IMDM medium containing 10% FBS and 1% penicillin-streptomycin. U937, THP-1, MOLM-13, NB-4, ML-2, Jurkat and RH30 cell lines were maintained in RPMI1640 medium containing 10% FBS and 1% penicillin-streptomycin. Cells were maintained in a humidified incubator at 37°C, 5% CO_2.

METHOD DETAILS

Murine and human MLL-AF9 leukemia model

BM cells were obtained from femoral and tibial bones, and stained with Lineage-Biotin (559971, BD Biosciences), Sca-1-PE (122508, BioLegend), c-KIT-Alexa Fluor 647 (105818, BioLegend) antibodies and Streptoavidin-FITC (554060, BD Biosciences), then LSK cells were isolated with FACS Aria (BD Biosciences). LSK cells were cultured in RPMI medium 1640 supplemented with 10% FBS, 20 ng/ml rmSCF, 10 ng/ml rmIL3, 10 ng/ml rmIL6. After 24 h of cultivation, spin infection was performed by adding retrovirus carrying MLL-AF9-ires-Neo or MLL-AF9-ires-GFP in the media containing 8 μg/ml polybrene to LSK cells and centrifuging at 1,400 x g for 90 min. MLL-AF9 expressing LSK cells were serially re-plated with methylcellulose medium (STEMCELL Technologies, MethoCult M3234) containing the above cytokines for 3 weeks. To obtain the fully transformed leukemia cells, immortalized cells were transplanted into lethally irradiated (9.5 Gy) syngeneic recipient mice along with 5×10^5 bone marrow cells from wild-type mice. Human cord blood CD34+ cells were enriched by microbeads (Miltenyi Biotec) and MLL-AF9 transduced cells were transplanted to NOD/SCID mice. AML cells were harvested from bone marrow or spleen and cultured in IMDM medium supplemented with 20% FBS, beta-mercaptoethanol, 20 ng/ml rhIL6/rhTPO/rhSCF/rhGM-CSF, 10 ng/ml IL3.

shRNA

miR-30 shRNA retroviral vector library and TRIN miR-E retroviral vector were obtained from Scott Lowe's lab (Zuber et al., 2011). RT3GEPIR miR-E retroviral vector was obtained from the RNAi Core in MSKCC (Fellmann et al., 2013). Plat-E retroviral packaging cells were transiently transfected with retroviral vector plasmids using X-tremegene 9 (Roche), and culture supernatants containing retroviruses were collected 48 h after transfection. For the doxycycline-inducible TRIN miR-E shRNA system, we transduced MSCV-rtTA-pgk-hygro transgene to MLL-AF9-ires-GFP expressing MLL-AF9 leukemia cells with retrovirus. In miR-30 shRNA experiments, infected cells were analyzed by flow cytometer every 3 days from 2 days post-infection. In TRIN miR-E shRNA experiments, cells were selected with 1mg/ml G418 for 2 days from 24 h post-infection to enrich the infected cells. The cells were treated with doxycycline, and analyzed at 2 days post-doxycycline to obtain the percentage of shRNA expressing cells. Doxycycline treated cells were analyzed again after 24 h, and analyzed every 3 days until 9 days or 12 days post-doxycycline. For *in vitro* studies, 1 μg/ml doxycycline was treated for every experiment. RT3GEPIR miR-E vector infected MLL-AF9 leukemia cells were selected with 2.5 μg/ml puromycin, and transplanted to sublethally irradiated (6.0 Gy) syngenic recipient mice. After 1-week post-transplant, we administrated doxycycline to the recipient mice by both food-intake (625 mg/kg diet, Harlan) and drinking water (2 mg/ml doxycycline with 5% sucrose). For human gene knockdown, 293T cells were transiently transfected with lentiviral vector plasmid pLKO1-puro with both pPAX2 and pMD2 packaging plasmids, and culture supernatants containing lentiviruses were collected 48 h after transfection. We transduced lentiviral vector by spin infection. shRNA expressing cells were selected with 2.5 μg/ml puromycin for 24 h and maintained in media containing 0.5 μg/ml puromycin.

Isolation of Leukocytes

PB cells were obtained from the facial vein. BM cells were obtained from femoral and tibial bones by aspiration. Splenocytes were isolated by pressing the tissues through a cell strainer (BD). Erythrocytes were removed with PB Pharm Lyse Lysing Buffer (BD).

Apoptosis, Cell cycle and Cell Differentiation

Apoptosis was evaluated with Annexin V Apoptosis Detection Kit APC (eBioscience). Cell cycle stage was analyzed with either Click-iT EdU Imaging Kit (Thermo Fisher Scientific) or BrdU Flow Kit (BD). Cell differentiation was monitored by microscopy. Cells were centrifuged onto glass slides and stained with the Jorvet Dip Quick stain kit (Fisher).

Sequence Alignment

Amino acid sequence of *Homo sapiens* SETD1A (NP_055527) / SETD1B (NP_055863), *Mus musculus* SETD1A (NP_821172) / SETD1B (NP_001035488), Mesocricetus auratus SETD1A (XP_005064405), *Xenopus* tropicalis SETD1A (XP_004919748), Alligator mississippiensis SETD1A (XP_006262890), Danio rerio SETD1A (XP_001920852), Poecilia reticulate SETD1A (XP_008414548) were compared with Vector NTI (Invitrogen).

CRISPR

pLKO5.sgRNA.EFS.GFP (Addgene #57822) and pLKO5.sgRNA.EFS.tRFP657 (Addgene #57824) constructs were a gift from Benjamin Ebert (Heckl et al., 2014). pCW-Cas9 lentiviral vector (Addgene #50661) was a gift from Eric Lander & David Sabatini (Wang et al., 2014). We designed 3-4 individual guide RNA sequences for each domain by CRISPR Design Tool (http://crispr.mit.edu/). 293T cells were transiently transfected with lentiviral vector plasmids with both pPAX2 and pMD2 packaging plasmids, and culture supernatants containing retroviruses were collected 48 h after transfection. First, we infected the doxycycline-inducible Cas9 vector to MLL-AF9-ires-GFP expressing MLL-AF9 leukemia cells, and established the parental clone which can induce high expression level of Cas9 by doxycycline. Next, we infected sgRNA vector, and then performed the experiments with these cells from 2 days post-infection. GFP+ cell frequencies were measured by BD FACSCanto II or LSRFortessa (BD Biosciences). To evaluate the genome editing frequency, we performed SURVEYOR assays with Suveyor mutation detection kit (IDT). For deep-sequencing analysis of CRISPR-targeting sites, genomic DNA was amplified using the primers on Table S4. Amplified PCR products were purified by PCR purification kit (QIAGEN). Libraries were prepared and sequenced on Mi-Seq instrument (illumina). The reference amplicon, annotated coding sequence, and the paired end reads were then used with CRISPResso (version 0.8.4; https://github.com/lucapinello/CRISPResso), with a quality filter of 10 and sequence trimming via Trimmomatic (http://www.usadellab.org/cms/index.php?page=trimmomatic). Cas9-expressing normal hematopoietic stem/progenitor cells were isolated by cell sorting of BM LSK cells from Rosa26-Cas9 knockin mice, and LSK cells were cultured in StemSpan SFEM (STEMCELL Technologies) containing 100 ng/ml rmSCF/rmFLT3/rmTPO. After 24 hr of cultivation, we infected sgRNA vector by spin infection. sgRNA-expressing LSK cells were isolated by cell sorter and were plated with methylcellulose medium (STEMCELL Technologies, MethoCult GF M3434) for 10 days. For the CRISPR domain scan of *Ccnk* gene, 188 sgRNAs with a RFP reporter were used for competitive growth assay. We purified sgRNA plasmid vectors with 96-well Miniprep Kit (Millipore) and performed a competitive growth assay in 96-well plates.

cDNA mutagenesis and rescue experiment

Myc- and DDK-tagged human SETD1A (NM_014712) and SETD1B (NM_015048) cDNA were obtained from Origene and DDK-tagged human CCNK (NM_001099402) cDNA was obtained from GenScript. Site-directed mutagenesis was performed by inverse PCR, with 5′-phosphorylated primers and CloneAmp HiFi PCR premix (Clontech). Each sequence of mutated cDNA was verified by Sanger sequencing, and protein expression was confirmed by western blotting. For the cDNA rescue experiment, *SETD1A* or *SETD1B* cDNA was cloned into pMSCV-ires-GFP vector, and retroviruses were generated in Plat-E cells. cDNA expression vectors were transduced into rtTA-expressing MLL-AF9 leukemia cells or *Setd1a*^{flox} MLL-AF9 leukemia cells by retrovirus, and GFP positive cells were sorted by FACSAria.

Immunoprecipitation

293T cells were cultured in 6-well plates, and transfected with Xtremegene 9 transfection reagent. Cells were harvested at 72 hr post-transfection, and lysed with 1x cell lysis buffer (Cell signaling) containing proteinase inhibitors (Biotool). 400 μg of protein lysate was loaded on pre-washed FLAG M2 affinity gel, and incubated overnight. After three TBS washes, we eluted the FLAG-tagged protein with 40 μL of TBS containing 3x FLAG peptides (ApexBio). Binding proteins were analyzed by western blotting or MS. For FUCCI-expressing 293T cells, cells were sorted and lysed with 1x cell lysis buffer. Protein lysate was mixed with Rabbit IgG or CCNK antibody, and incubated overnight. The lysate was loaded on pre-washed Protein A/G Dynabeads, and incubated for 2 hr at 4°C, then eluted by 2x SDS sample buffer. Proteins were analyzed by western blotting.

Western blotting

Cells were lysed in 1x cell lysis buffer (Cell Signaling Technology) containing protease inhibitor cocktail (Bimake). Lysis was completed via ultrasonication and proteins were quantified with BCA Protein Assay Kit (Thermo Fisher Scientific). Lysate was mixed with 4xLDS sample buffer (Invitrogen) and beta-mercaptoethanol, and proteins were denatured by boiling. Denatured proteins were separated on a 4%–12% Bis-Tris gel (ThermoFisher) in NuPAGE MOPS SDS running buffer (Thermo Fisher Scientific) and transferred to a PVDF membrane (Millipore) in NuPAGE Transfer Buffer (Thermo Fisher Scientific) with XCell II Blot Module (Thermo Fisher Scientific). Blots were blocked with 5% skim milk in TBST for 30 min, and incubated with primary antibodies (which are listed in the Key Resources Table) at 4°C overnight. Immunocomplexes were labeled by HRP-conjugated anti-mouse IgG or anti-rabbit IgG and visualized using Femto Maximum Sensitivity Substrate (Thermo Fisher Scientific). The signals were deteceted with ImageQuant LAS 4000 (GE Healthcare) and signal intensities were quantified using ImageJ software.

ChIP and RNA analyses

For ChIP-seq, *Setd1a*$^{\Delta/\Delta}$ and *Setd1a*$^{+/+}$ leukemia cells were analyzed at day 3 post-tamoxifen. shRNA-expressing NIH 3T3 cells were analyzed at day 4 post-doxycycline. Cells were fixed with 2mM DSG for 30 min followed by 1% formaldehyde for 10 min and treated with 0.125M Glycine for 5 min. Fixed cells were washed twice with cold PBS. Washed cells were suspended in ChIP Lysis Buffer and shredded using Bioruptor (Diagenode) or E220 Focused-ultrasonicator (Covaris). Appropriate amounts of antibodies were added into the chromatin and incubated overnight at 4°C. The immune complex was collected with protein A/G agarose or magnetic beads and washed sequentially in the low salt wash buffer (20mM Tris pH8, 150mM NaCl, 0.1% SDS, 1% Triton X-100, 2mM EDTA), the high salt wash buffer (20mM Tris pH8, 500mM NaCl, 0.1% SDS, 1% Triton X-100, 2mM EDTA), the LiCl wash buffer (10mM Tris pH8, 250mM LiCl, 1% NP-40, 1% Sodium Dexyxholate, 1mM EDTA) and TE. Chromatin was eluted with elution buffer (1% SDS, 0.1 M NaHCO3), and then reverse cross-linked with 0.2M NaCl at 65°C for 4 hr. DNA was purified with a PCR purification kit (QIAGEN) or by PC8 extraction and isopropanol precipitation. For RNA-seq, *Setd1a*$^{\Delta/\Delta}$ and *Setd1a*$^{+/+}$ leukemia cells were analyzed at day 4 post-tamoxifen. *Ccnk* sgRNA-expressing cells were analyzed at day 4 post-doxycycline. For RNA, total RNA was purified with RNeasy Mini kit (QIAGEN). cDNA was synthesized with Tetro cDNA synthesis kit (Bioline). cDNA fragments were quantified by a TaqMan Gene expression assay (Applied Biosystems). ChIP DNA or cDNA fragments were quantified by SYBR Green Real-time PCR or a TaqMan Gene expression assay (Applied Biosystems) with a ViiA 7 Real-time PCR system. Libraries were prepared by using a ThruPLEX DNA-seq Kit (Rubicon Genomics). The DNA library was validated using TapeStation (Agilent Technologies) and was quantified using a Qubit 2.0 Fluorometer (Thermo Fisher Scientific). Libraries were pooled and sequenced on Illumina HiSeq2000 or Illumina NextSeq500. Data were analyzed by workflows on Basepair (https://www.basepairtech.com). Heatmaps of ChIP-seq data were generated by using EaSeq software (http://easeq.net).

FUCCI

ES-FUCCI (Citrine-Geminin(1-110)/mCherry-hCdt1(30-120)) was a gift from Pierre Neveu (Addgene plasmid #62451) (Sladitschek and Neveu, 2015). We transduced the vector into the 293T cell line and established single cell-clones that stably express the reporters. Cells in each cell cycle stage were sorted by FACSAria.

MS analysis

Unless otherwise specified, all chemicals were from Sigma-Aldrich at the highest available purity. Proteins were separated with gradient Bis-Tris Protein Gel (Thermo Fisher Scientific) and stained with Pierce Silver Stain for Mass Spectrometry kit (Thermo Fisher Scientific). Lanes containing PAGE-resolved proteins were excised from the gel and divided into eight pieces, that were de-stained using 30 mM potassium hexa-cyanoferrate (III) /100 mM sodium thiosulfate, washed with 25 mM ammonium bicarbonate pH8.4 / 25% (v/v) Optima LC/MS acetonitrile (Thermo Fisher Scientific, Fair Lawn, NJ) and de-hydrated using a vacuum centrifuge. Proteins were reduced (10 mM DTT / 100 mM ammonium bicarbonate pH8.4, 56°C for 60 min) and alkylated (55 mM iodoacetamide / 100 mM ammonium bicarbonate pH8.4, 25°C for 30 min, dark), and excess iodoacetamide was removed by three cycles of dehydration (100 μl C_2H_3N) and rehydration (100 μl of 100 mM ammonium bicarbonate pH8.4) prior to final dehydration in a vacuum centrifuge. Gel slabs were re-hydrated using 50 mM ammonium bicarbonate pH8.4 containing 0.04 μg sequencing grade modified porcine trypsin (Promega, Madison, WI) per reaction and proteolysis was allowed to proceed for 16 hr at 37°C. Proteolysis inhibition and tryptic peptide elution from the polyacrylamide slabs was achieved by two consecutive 30 min incubations in 5% formic acid/ 70% acetonitrile (v/v) under continuous shaking. Eluates from each sample were pooled, lyophilized and stored at −80°C until analysis.

Peptide pellets were resuspended in 0.1% formic acid (99+%, Thermo Fisher Scientific, Rockford, IL) / 3% acetonitrile and 5% of the solution was analyzed by LC/MS. The LC system consisted in a vented trap-elute setup (Ekspert nanoLC 425, Eksigent, Redwood city, CA) coupled to the Orbitrap Fusion mass spectrometer (Thermo Fisher Scientific, San Jose, CA) via an nano electro-spray DPV-565 PicoView ion source (New Objective, Woburn, MA). The trap column was fabricated capping a 5 cm × 150 μm internal diameter silica capillary (Polymicro Technologies, Phoenix, AZ) with a 2 mm silicate frit, and pressure loaded with Poros R2-C18 10 μm particles (Life Technologies, Norwalk, CT). The analytical column consisted of a 25 cm × 75 μm internal diameter column with integrated electrospray emitter (New Objective, Woburn, MA), and was packed with ReproSil-Pur C18-AQ 1.9 μm particles (Dr. Maisch, Ammerbuch-Entringen, Germany). Samples were loaded on the trap column at 1 μl/minute with analytical column excluded from the flow path. The analytical column was then put in-line with the trap, and peptides were resolved over 120 min using a 5%–40% gradient acetonitrile/ 0.1% formic acid (buffer B) gradient in water/0.1% formic acid (buffer A) at 300 nl/minute. Precursor ion scans were recorded from 400–2000 m/z in the Orbitrap (120,000 resolution at m/z 200) with automatic gain control target set 10^5 ions and maximum injection time of 50 ms. We used data-dependent mass spectral acquisition with monoisotopic precursor selection, charge ion selection (2–7), dynamic exclusion (60 s, 5 ppm tolerance), HCD fragmentation (normalized collision energy 32, isolation window 1 Th) using the top speed algorithm with a duty cycle of 2 s. Product ion spectra were recorded in the linear ion trap (“normal” scan rate, automatic gain control = 10^4 ions, maximum injection time = 150 ms). Raw files are submitted to the MassIVE repository with accession number MSV000080012.

Raw files were submitted to MaxQuant 1.4.1.2 to be searched against the human UniProt database (version 04/22-14, containing isoforms) with FDR < 0.01 at both peptide and protein level. Mass tolerance was set at 5 ppm for orbitrap spectra and 0.5 Da for linear

trap spectra. C-carbamidomethylation was set as fixed modification, while phosphorylation of S/T/Y and M-oxidation were allowed as variable. The software was set to align chromatographic features (Matching time window = 1 min, Alignment time window = 20 min). Identified proteins mapping in the common contaminants database cRAP were removed from the output. GO term analysis was performed with GOTermFinder (http://go.princeton.edu/cgi-bin/GOTermFinder).

THZ531, CDK12/13 covalent inhibitor

The half-maximal inhibitory of THZ531 in 293T-FUCCI cells was determined (IC_{50} = 60 nM). 100 nM THZ531 was treated for 72 hr and cells from each cell cycle stage were collected by cell sorter.

QUANTIFICATION AND STATISTICAL ANALYSIS

Error bars in all of the data represent a standard deviation. Number of replicates and number of repeated experiments are reported in the Figure legends. For statistical comparison, we performed a Student's t test or one-way ANOVA followed by Tukey's test. Data with statistical significance (*$p < 0.05$, **$p < 0.01$) are shown in figures. Statistical analyses were performed using Prism 7 software (GraphPad).

DATA AND SOFTWARE AVAILABILITY

The accession number for the RNA-seq and ChIP-seq data reported in this paper is NCBI GEO: GSE108615. The accession number for the MS analysis data reported in this paper is MassIVE: MSV000080012. The accession number for the raw data reported in this paper is Mendeley Data: https://doi.org/10.17632/hhjhsrgkz7.2.

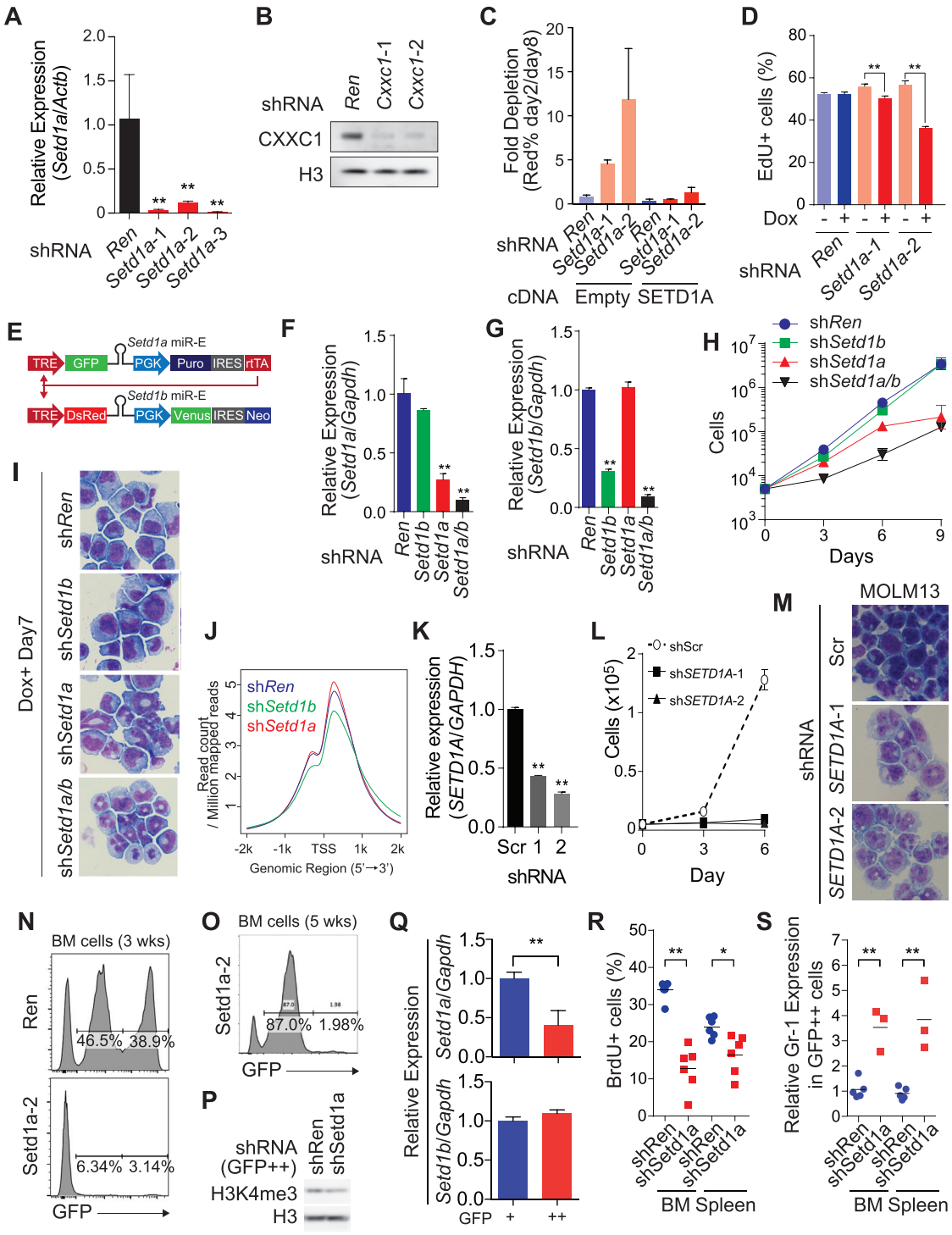


Figure S1. Knockdown of Setd1b by miR-E Suppresses Cell Growth, but Does Not Induce Cell Differentiation in MLL-AF9 Leukemia Cells, Related to Figure 1

(A) *Setd1a* expression was analyzed by qRT-PCR in mouse MLL-AF9 leukemia cells that were transfected with non-targeting control (sh*Ren*) or 3 independent *Setd1a* shRNAs. Representative data of one out of 3 independent experiments are shown.

(B) CXXC1 protein levels were analyzed by western blot in mouse MLL-AF9 leukemia cells that were transfected with non-targeting control (sh*Ren*) or 2 independent *Cxxc1* shRNAs. Representative data of one out of 2 independent experiments are shown.

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(C) Mouse MLL-AF9 leukemia cells were transfected with a human SETD1A construct with a GFP reporter and mouse *Setd1a* shRNA (*Setd1a-1*, *Setd1a-2*) with a DsRed reporter. DsRed percentages in GFP-positive cells were assessed at 2 days and 8 days post shRNA-transduction. Negative selection is plotted as the fold change of DsRed percentage. Representative data from one out of 2 independent experiments are shown.
(D) Cell cycle stage was analyzed in *Setd1a* shRNA-expressing mouse MLL-AF9 cells. Mouse MLL-AF9 leukemia cells harboring dox-inducible shRNA vector were analyzed after 4 days of dox treatment. EdU was treated for 2 hr and EdU-incorporated cells were counted by flow-cytometry. This experiment was repeated 3 times with 3 biological replicates in each experiment.
(E) Schematics are shown of the Dox-inducible shRNA vectors used for *Setd1a* and *Setd1b* double knockdown. rtTA is expressed from one shRNA vector and the DsRed reporter can be expressed from the other shRNA vector.
(F and G) Relative expression levels of *Setd1a* (F) or *Setd1b* (G) were analyzed in double knockdown cells. Mouse MLL-AF9 leukemia cells harboring two shRNA vectors were selected by puromycin and G418, and cells were analyzed at 4 days post-dox.
(H) Numbers of *Setd1a/b* shRNA-expressing mouse MLL-AF9 leukemia cells were counted every 3 days after dox treatment. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.
(I) *Setd1a/b* shRNA-expressing mouse MLL-AF9 leukemia cells were stained with May-Grünwald/Giemsa at 7 days post-dox. Representative images from 2 independent experiments with 3 biological replicates are shown.
(J) ChIP-seq analysis was performed for H3K4me3 in *Setd1a* or *Setd1b* shRNA-expressing mouse MLL-AF9 leukemia cells at 4 days post-dox. Average profiles of H3K4me3 are shown and focus 2 kbp upstream and downstream of the TSS.
(K) *Setd1a* expression was analyzed by qRT-PCR in MOLM-13 cells that were transfected with scramble control (shScr) or two independent *Setd1a* shRNAs (sh*SETD1A*-1 and sh*SETD1A*-2). Cells were harvested at 4 days post-transfection. This experiment was repeated 3 times with 3 biological replicates in each experiment.
(L) Numbers of *SETD1A* shRNA-expressing MOLM-13 cells were counted every 3 days. Representative data of one out of 2 independent experiments with 3 biological replicates are shown.
(M) *SETD1A* shRNA-expressing MOLM-13 cells were stained with May-Grünwald/Giemsa after 4 days of dox treatment. Representative images from 2 independent experiments with 3 biological replicates are shown.
(N) Histograms are shown of GFP in BM cells of *Ren* or *Setd1a* shRNA-expressing leukemia mice at 3 weeks post-transplant. Mouse MLL-AF9 leukemia cells harboring dim GFP were transfected with shRNA vector with GFP. Dox induction produces the cell population with bright GFP.
(O) Histogram is shown of GFP in BM cells of *Setd1a* shRNA-expressing leukemia mice at 5 weeks post-transplant. Dox-unresponsive leukemia cells occupied the BM. Representative image is shown of 5 independent recipient mice.
(P) Global H3K4me3 levels were analyzed by western blot in *Setd1a* shRNA-expressing leukemia cells from the spleen.
(Q) *Setd1a* (upper) and *Setd1b* (bottom) expression levels were analyzed by qRT-PCR in GFP+ (shRNA-) and GFP++ (shRNA^{+}) cells from recipient mice harboring *Setd1a* shRNA-expressing leukemia cells at 3 weeks post-transplant.
(R) Cell cycle stage was analyzed in *Setd1a* shRNA-expressing leukemia cells from the BM and spleen. BrdU was injected into mice 2 hr prior to sacrifice. Sorted GFP++ leukemia cells were stained with anti-BrdU antibody and analyzed by flow cytometry. (n = 6 / group).
(S) Relative intensity of Gr-1 (granulocytic differentiation marker) was analyzed in *Setd1a* shRNA-expressing leukemia cells from the BM and spleen. (n = 3-5 / group).

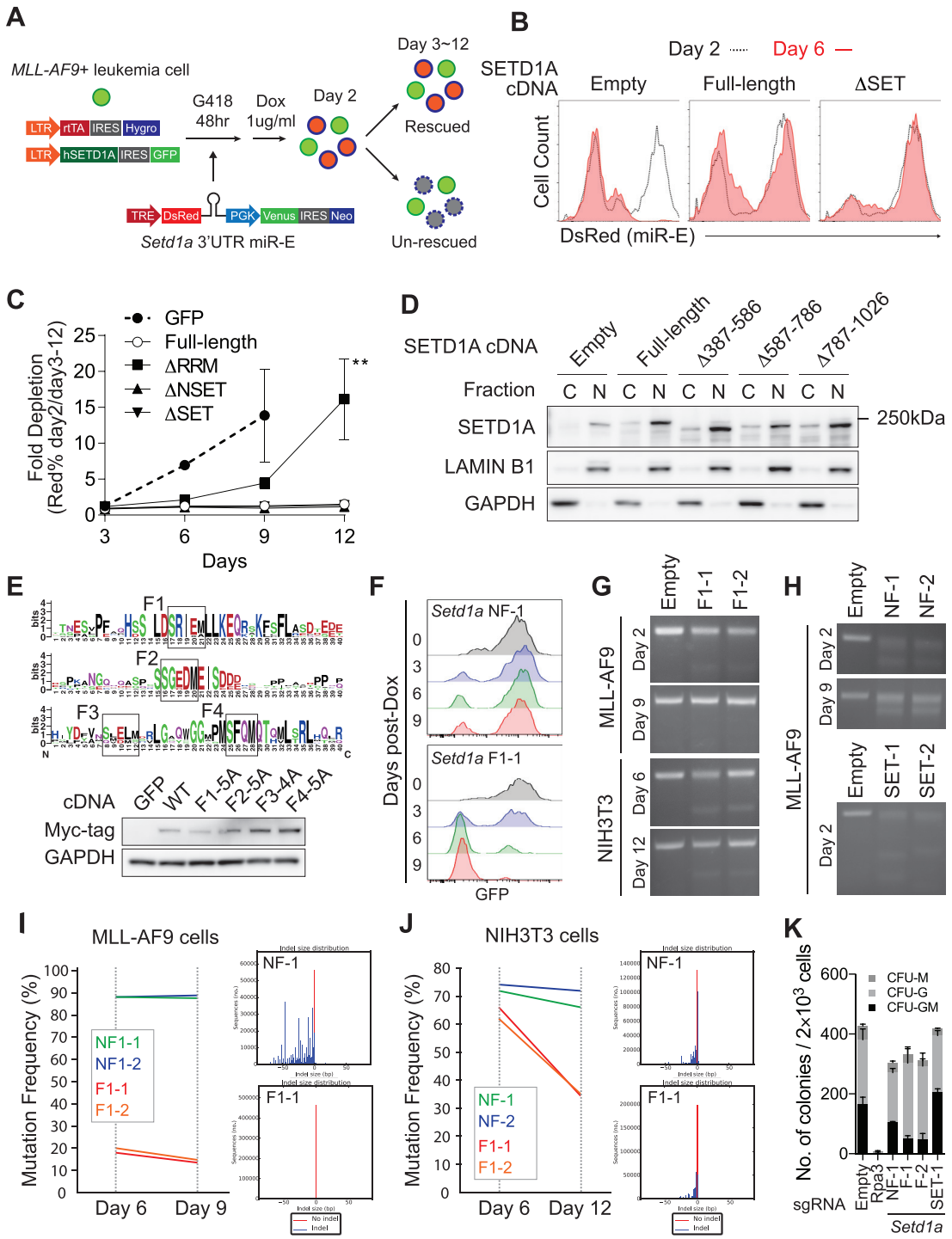


Figure S2. Internal Region on SETD1A Is Required for Leukemia Growth, but Not Protein Stability and Protein Nuclear Localization, Related to Figure 2

(A) Scheme depicts the shRNA-based SETD1A rescue experiment. Two individual shRNAs targeting mouse *Setd1a* 3′UTR were used for the knockdown.

(B) Function of SETD1A mutants was evaluated by number of DsRed positive cells. Representative histograms of flow cytometric analysis in *SETD1A* cDNA rescue study at 2 days (dot line) and 6 days (red) post-dox induction of *Setd1a* shRNAs are shown.

(C) The fold depletion is graphed of RRM, NSET, or SET domain-deleted SETD1A expressing cells after the induction of *Setd1a* shRNAs. GFP-expressing empty vector was used as a control.

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(D) Western blot analysis was used to determine the protein expression levels of the non-functional SETD1A mutants in the cytoplasm and nucleus. Proteins were extracted from 293T cells which were transiently expressing each construct. LAMIN B1 and GAPDH were used as markers for nuclear (N) and cytoplasmic (C) fractions, respectively.

(E) Evolutionarily conserved sequences of FLOS1–4 (F1–4) in SETD1A are depicted. Bottom panel shows the protein expression levels of the alanine substituted mutants of FLOS1–4.

(F) The indicated *Setd1a* sgRNAs with a GFP reporter were transfected and GFP was monitored by flow cytometry every 3 days. Dox was treated after 2 days of transfection. Representative histograms depicting GFP positive cells are shown.

(G and H) SURVEYOR assay was performed to check for edited alleles in NIH 3T3 and MLL-AF9 leukemia cells after doxycycline treatment. FLOS1 was edited by F1-1 and F1-2 sgRNAs (G). The non-functional region and SET domain were edited by NF1-1/2 and SET-1/2 sgRNAs, respectively (H).

(I and J) Deep-sequencing analysis of sgRNA-expressing MLL-AF9 leukemia cells (I) and NIH 3T3 cells (J) was performed. Left panel shows mutation frequency at the indicated days post-dox. Right panels show indel size distribution in NF-1 (top) and F1-1 (bottom) sgRNA-infected samples.

(K) Colony forming ability of Cas9-expressing normal hematopoietic stem/progenitor cells with *Setd1a* sgRNAs. Representative data from 2 independent experiments with 3 biological replicates are shown.

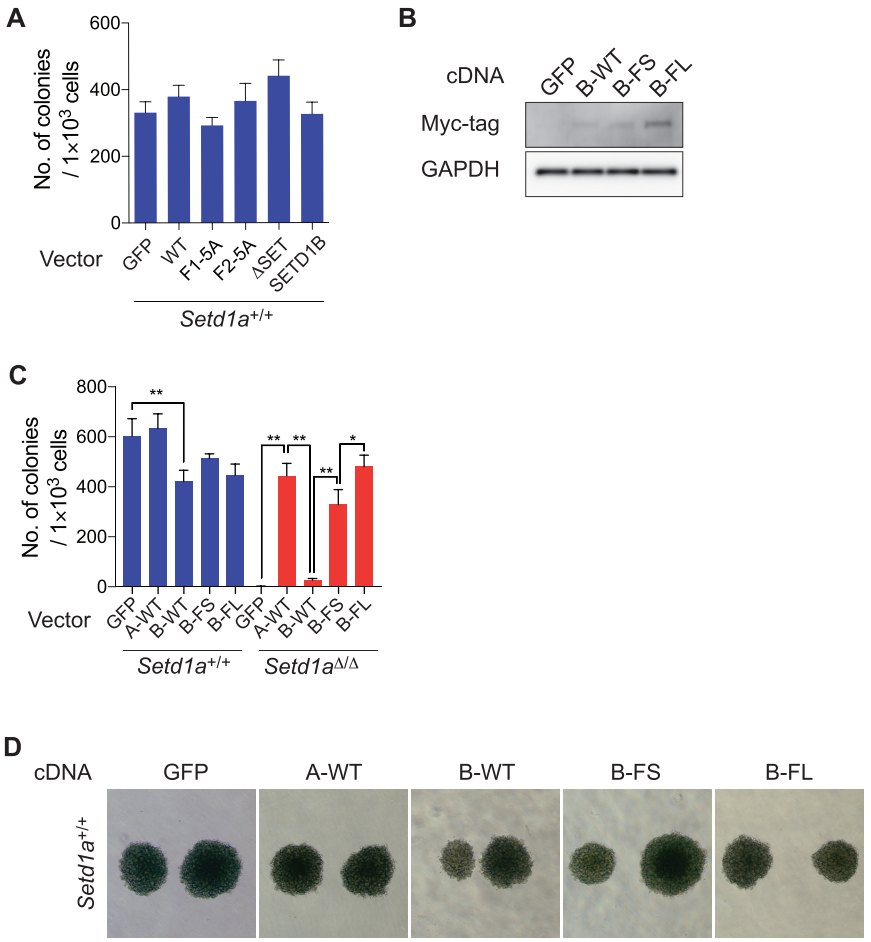


Figure S3. Known Domains of SETD1A/B Are Interchangeable for Leukemia Cell Growth, Related to Figure 3

(A) *Setd1a*$^{+/+}$;*CreER* cells were transfected with SETD1A or SETD1B constructs. The cells were treated with tamoxifen, then assayed for colony-forming potential. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

(B) Western blot analysis was performed using SETD1 chimeric mutant-expressing 293T cells. These constructs were used for the rescue experiments. Representative images from 3 independent experiments are shown.

(C) *Setd1a*$^{+/+}$;*CreER* or *Setd1a*$^{fl/fl}$;*CreER* cells were transfected with SETD1 chimeric constructs. The cells were treated with tamoxifen, then assayed for colony-forming potential. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

(D) Colony forming ability of *Setd1a*$^{+/+}$;*CreER* cells expressing SETD1 chimeric constructs was evaluated with methylcellulose. Representative images of colonies from 3 independent experiments are shown.

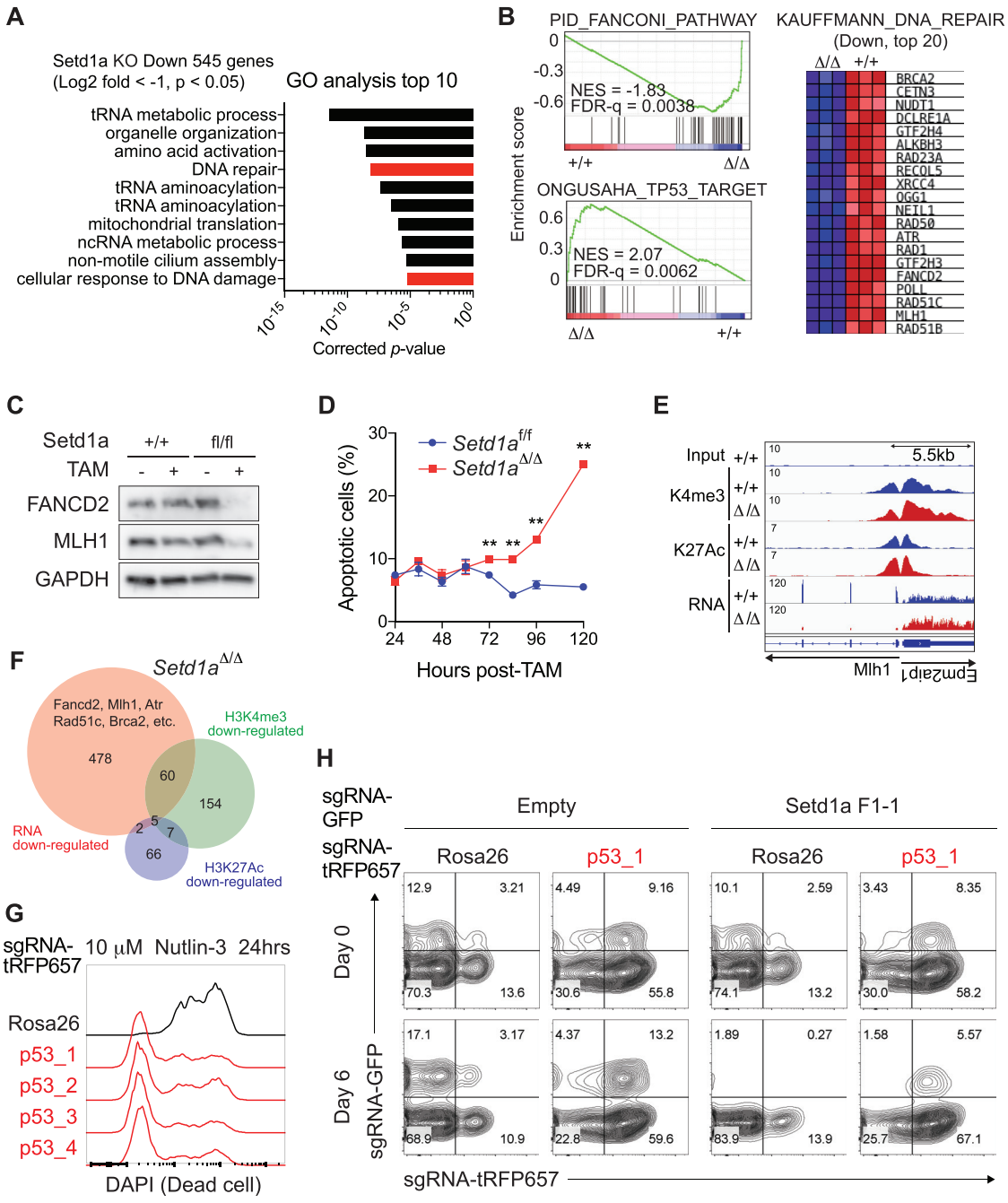


Figure S4. SETD1A FLOS Domain Regulates DNA Damage Response, Related to Figure 4

(A) Gene ontology (GO) analysis was applied to the 545 downregulated genes (Log_2 fold change < −1, p < 0.05) in *Setd1a*-deficient leukemia cells. The bars for DNA damage stimulus are shown in red.

(B) RNA-seq data from *Setd1a*$^{\Delta/\Delta}$ and *Setd1a*$^{+/+}$ leukemia cells were analyzed for gene set enrichment. Representative enrichment plots from the fanconi pathway and p53 target gene set are shown (left). The list of 20 genes downregulated by *Setd1a* disruption in the KAUFFMANN DNA REPAIR gene set is also shown (right).

(C) Western blot analysis revealed protein levels of FANCD2 and MLH1 in *Setd1a*-deficient MLL-AF9 leukemia cells at 4 days post-tamoxifen. Representative images from 2 independent experiments are shown.

(D) Annexin V$^+$DAPI$^-$ population in *Setd1a*$^{fl/fl}$;*CreER* MLL-AF9 leukemia cells was analyzed from 24–120 hr post-tamoxifen. This experiment was repeated 2 times with 3 biological replicates in each experiment.

(E) Histograms depict H3K4me3, H3K27Ac and mRNA peaks on the *Mlh1* locus in *Setd1a* knockout leukemia cells. For ChIP-seq, *Setd1a*$^{\Delta/\Delta}$ and *Setd1a*$^{+/+}$ leukemia cells were analyzed at day 3 post-tamoxifen.

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(F) The overlap between RNA downregulated genes, H3K4me3 downregulated genes and H3K27Ac downregulated genes in *Setd1a* knockout cells are shown. Decreasing of H3K4me3 and H3K27Ac were determined using edgeR.
(G) *p53* sgRNA-expressing leukemia cells are resistant to Nutlin-3 (p53 activator) treatment. Nutlin-3 (10 μM, 24 hr) treated cells were stained with DAPI to monitor the dead cell populations.
(H) Representative cytograms depict dual sgRNA expressing cells. iCas9-expressing mouse MLL-AF9 leukemia cells were transfected with *Rosa26* and *p53* sgRNA with a tRFP657 reporter and treated with dox for 2 days. After a washout of dox, empty and *Setd1a* F1-1 sgRNA with a GFP reporter were transfected. Dox was re-treated at day 2 post-infection. Cells were analyzed at day 0 and day 6 post-dox. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

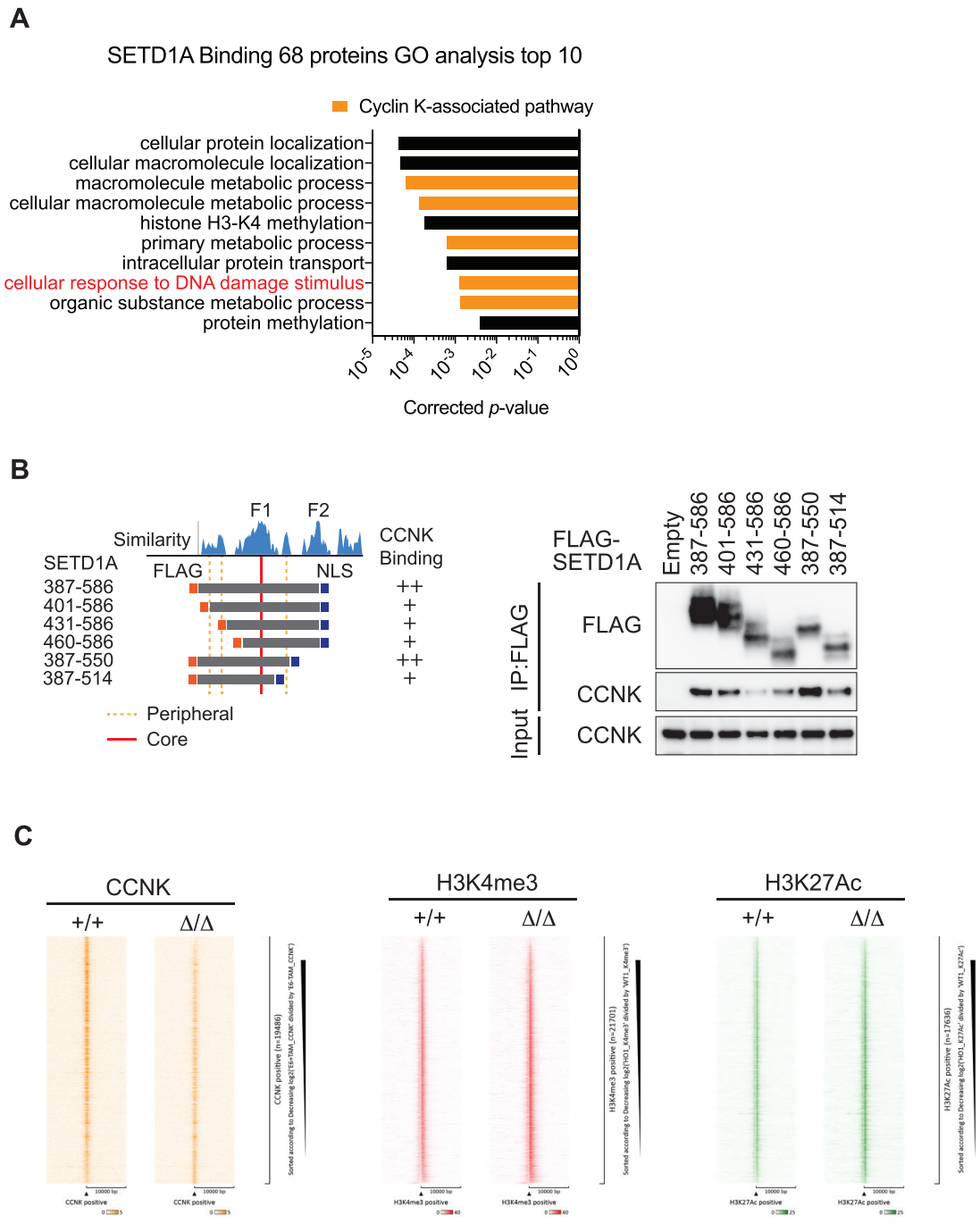


Figure S5. Cyclin K Is a SETD1A-Binding Protein, Related to Figure 5

(A) 68 SETD1A-binding proteins were used for GO analysis. The bar for DNA damage stimulus is shown in red, and the gene ontology groups including cyclin K are shown in orange.

(B) FLAG-tagged fragments spanning the 387–586 aa in SETD1A were used for Co-IP with CCNK. Illustrations of the FLAG-tagged fragments are on the left. Representative images from one out of 2 independent experiments are shown.

(C) Density of ChIP-seq reads for CCNK, H3K4me3 and H3K27Ac of TSS from *Setd1a*$^{+/+}$ and *Setd1a*$^{\Delta/\Delta}$ leukemia cells. Regions are ranked according to their log2 fold change between *Setd1a*$^{+/+}$ and *Setd1a*$^{\Delta/\Delta}$ cells.

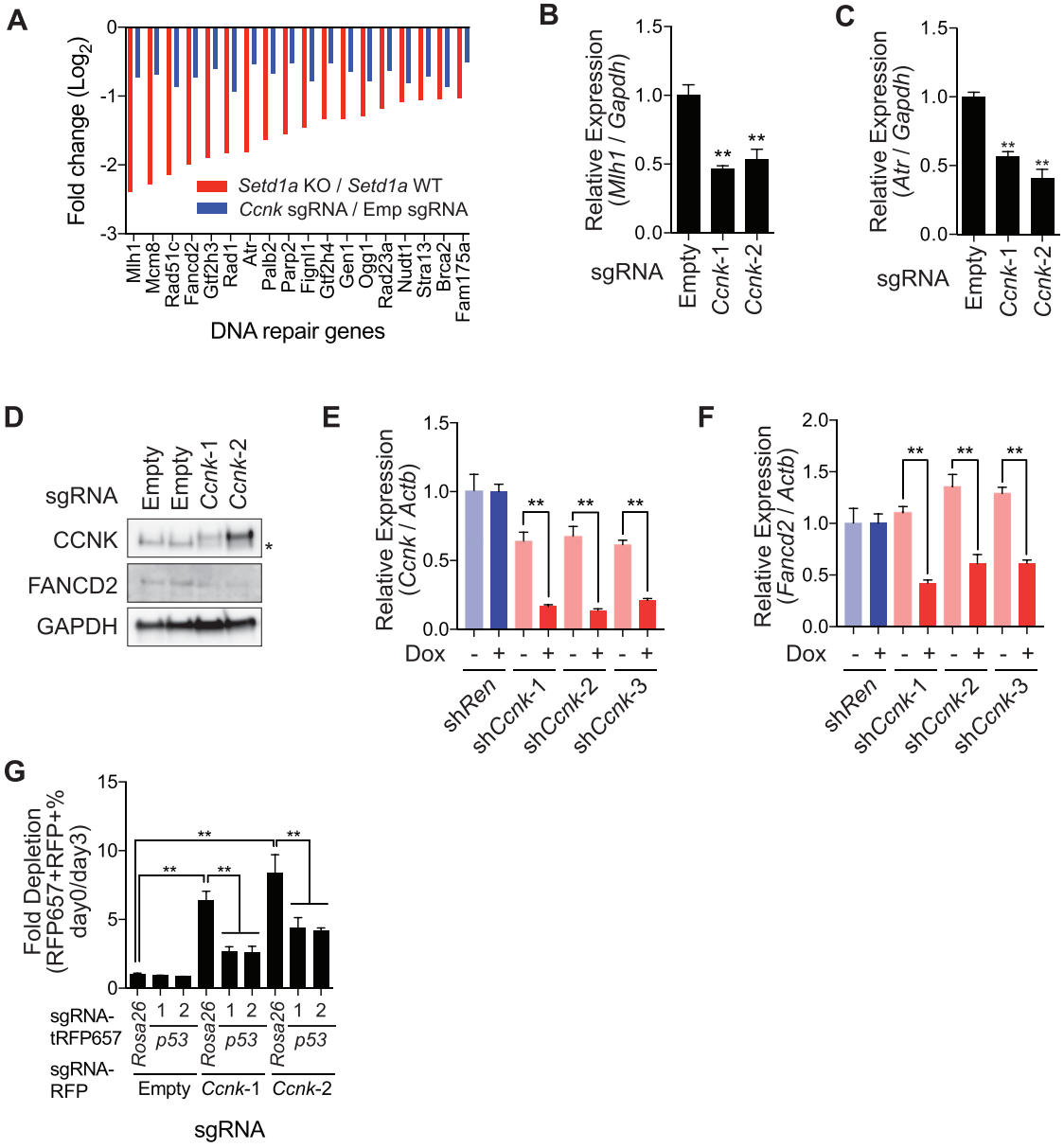


Figure S6. Cyclin K Regulates DNA Damage Response Genes in MLL-r Leukemia, Related to Figure 6

(A) Fold changes are shown of the 18 DNA repair pathway-associated genes that are classified in GO analysis shown in Figure 6D.

(B and C) *Mlh1* (B) and *Atr* (C) expression levels were analyzed by qRT-PCR in *Ccnk* sgRNA-expressing MLL-AF9 leukemia cells. RFP positive cells were sorted and total RNA was extracted at 4 days post-dox. Representative data from one out of 2 independent experiments are shown.

(D) Western blots show representative Protein expression of CCNK and FANCD2 in *Ccnk* sgRNA-expressing MLL-AF9 leukemia cells. RFP positive cells were sorted and total protein was extracted at 4 days post-dox. Asterisk in image indicates the endogenous CCNK protein bands. Representative images from one out of 2 independent experiments are shown.

(E and F) *Ccnk* (E) and *Fancd2* (F) expression were analyzed in mouse MLL-AF9 leukemia cells that were transfected with doxycycline (dox)-inducible control (shRen) or *Ccnk* shRNAs. Cells were analyzed at 24 hr post-dox. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

(G) *p53* sgRNAs with a tRFP657 reporter and *Ccnk* sgRNAs with a RFP reporter were sequentially transfected into iCas9-expressing MLL-AF9 leukemia cells. The fold depletion of double sgRNA positive cells was measured at 3 days. Empty sgRNA vector or *Rosa26* sgRNA vector were used as negative controls. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

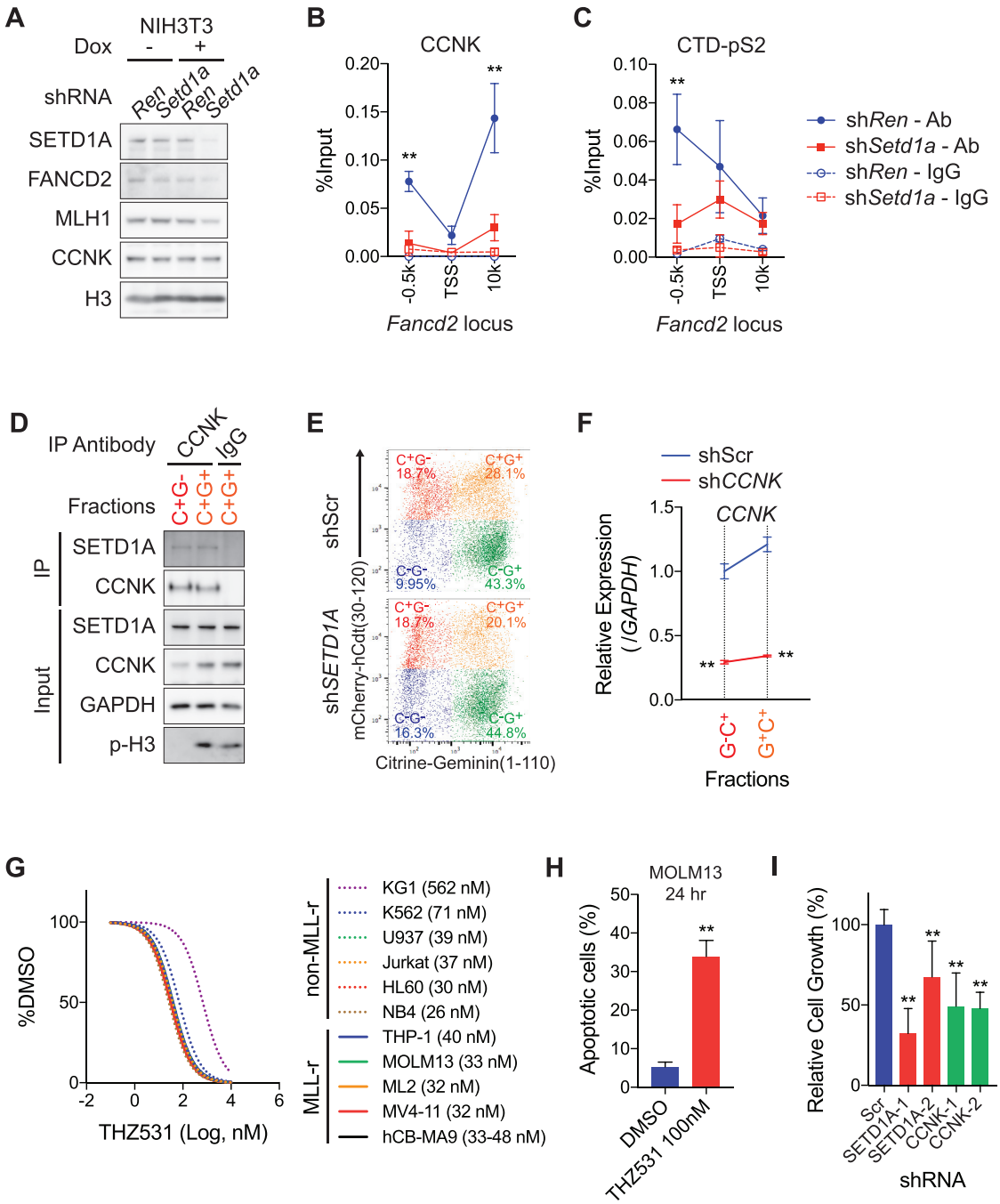


Figure S7. SETD1A-Cyclin K Regulates *FANCD2* Transcription during the Cell Cycle, Related to Figure 7

(A) Western blot analysis determined *Setd1a*-knockdown efficiency in NIH 3T3 cells. Total protein was extracted at 4 days post-dox. Representative images from 2 independent experiments are shown.

(B and C) ChIP-qPCR was performed for cyclin K (B) and RNAP2 CTD-pSer2 (C) on the *Fancd2* locus in *Ren* shRNA (blue) or *Setd1a* shRNA (red) -expressing NIH 3T3 cells at 4 days post-dox. Rabbit IgG was used as negative control in ChIP (dot lines). Representative data from 3 independent experiments with 3 biological replicates are shown.

(D) Native cyclin K Co-IP experiments were performed with fractioned 293T-FUCCI cells. Total protein was extracted from sorted cells and IP was performed with cyclin K antibody or IgG. Representative images from 2 independent experiments are shown.

(E) Representative cytograms depict the cell cycle stage distribution of Scramble shRNA (shScr) or *SETD1A* shRNA (sh*SETD1A*) –expressing 293T-FUCCI cells. Cells were analyzed at day 4 post-shRNA transduction.

(F) *CCNK* expression was analyzed by qRT-PCR in *CCNK* shRNA-expressing 293T-FUCCI cells. Scramble shRNA (blue) and *CCNK* shRNA (red) –expressing cells were analyzed at day 4 post-transfection. Representative data from one out of 2 independent experiments with 3 biological replicates are shown.

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(G) MLL-r/non-MLL-r leukemia cell lines and primary human AML cells generated by MLL-AF9 transduced cord blood cells were treated with THZ531 for 72 hr and assessed for anti-proliferative effects by cell counting. 6 independent MLL-AF9 transduced human cord blood cells (hCB-MA9) are used.
(H) Annexin V+DAPI- population in THZ531-treated MOLM-13 cells was analyzed at 24 hr post-treatment. This experiment was repeated 2 times with 3 biological replicates in each experiment.
(I) Numbers of *SETD1A* or *CCNK* shRNA-expressing primary human AML cells were counted at day 2 and day 8 post-infection. Data from 2 independent AML samples with 3 biological replicates are shown.